Introduction to PHYS5033
&
Structure of input-output tables

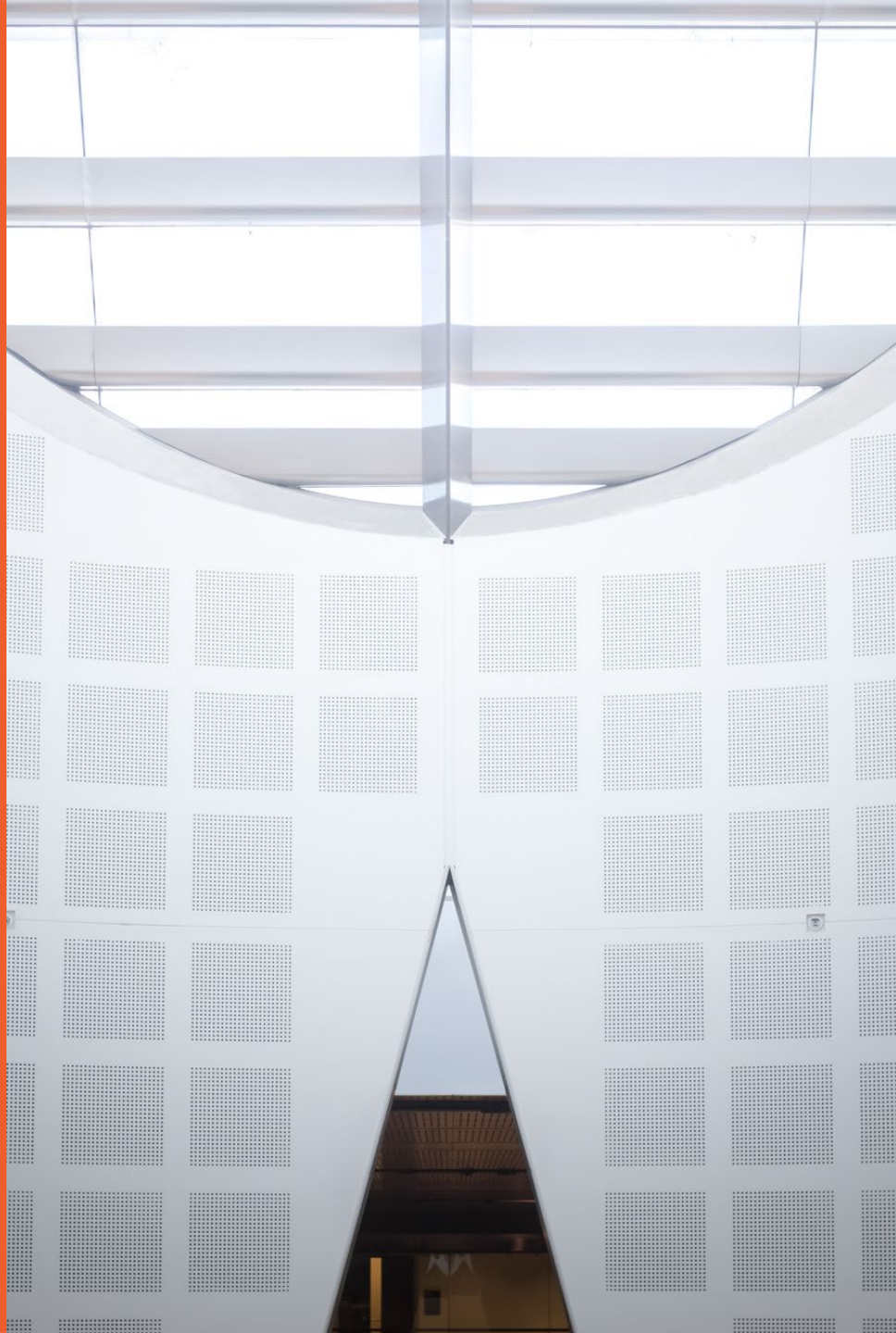
Lecturer: Dr Amanda Irwin

Unit coordinator: Prof. Arunima Malik

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arunima.malik@sydney.edu.au

THE UNIVERSITY OF SYDNEY



We recognise and pay respect to the Elders and communities – past, present, and emerging – of the lands that the University of Sydney's campuses stand on. For thousands of years they have shared and exchanged knowledges across innumerable generations for the benefit of all.



Welcome and information to support your University journey

THE UNIVERSITY OF SYDNEY

CRICOS 00026A



Emergency and evacuation

In the unlikely event of a serious emergency, such as a seriously ill or injured person or threat, call triple zero (000) for emergency services.

During an emergency, you may be required to evacuate. All University staff, students and visitors are required to **respond to emergency alarms** and **follow instructions**.

- **Check** for any sign of immediate danger
- **Shutdown** equipment/processes that cannot be left unattended
- **Locate** your closest exit
- **Follow** the green exit signs to find your way out of the building **(do not use lifts)**
- **Proceed** to the assembly area, a safe distance from the building, while waiting until the emergency is over.

More information on staying safe in the event of an emergency is available at:
www.sydney.edu.au/about-us/campuses/emergencies-and-personal-safety.html

Emergency lockdown or shelter in place

In some circumstances, we might be asked to remain inside the building for our own safety. We call this a lockdown or shelter-in-place.



You will be notified of an emergency lockdown via **building announcements or other mechanisms** including **verbally by Protective Services or Building Emergency Wardens.**



It is important that you **follow directions from a Warden, Protective Services and/or Emergency Services,** remain calm and **stay inside the building.**



If your life is being threatened, or if someone is seriously ill or injured, or if you just witnessed a serious incident **dial triple zero (000) for emergency services** (fire/ambulance/police).

More information on staying safe in the event of an emergency is available at:
www.sydney.edu.au/about-us/campuses/emergencies-and-personal-safety.html

Safezone App

SafeZone is a **free safety app** for all University of Sydney students and staff. It connects you quickly with Protective Services, emergency teams, and sends real-time safety alerts.

Get started with SafeZone by **downloading the app** and logging in with your **university email.**

While on campus, you can access features including:

- **Emergency assistance:** Connect instantly with Protective Services
- **First aid:** Request medical help
- **Timed Check-In:** Set a timed check-in when working or studying alone
- **Call Security:** If you're locked out, lost on campus or need an escort
- **Report something:** If you notice something concerning or unusual
- **Wellbeing support:** Connect with student support services
- **Location sharing:** So help can reach you faster
- **Push notification alerts:** Keep informed about critical incidents



More information on SafeZone is available at: sydney.edu.au/safezone

Your role in ensuring a safe campus

We are committed to providing a safe and healthy learning environment that is free from bullying, unlawful harassment and discrimination.



You can ensure the safety of yourself and your peers by understanding your rights and responsibilities, which are detailed in our Student Charter.



If you experience or witness any form of behaviour that breaches the Student Charter or our codes of conduct, you can report it to our Student Affairs team.



Our support services, including the Wellbeing team and Safer Communities Office, can support you in dealing with any threats to your safety. You can connect with these teams via the Student Portal.

https://www.sydney.edu.au/students/complaints.html

Accessing tools and support

Accessing tools and support for your studies and wellbeing

Stay Organised. Stay Connected. Stay Supported.
Make uni life easier with the Student Portal & Sydney Uni App

Stay organised – Use the new **Study Dashboard** to plan for your semester and view your assignments and to-do lists all in one place.

Stay supported – Need study help, wellbeing resources, or student services? The **Service and Support page** has everything you need

Stay connected – Find events on **What's On**, track your classes and exams with the **Calendar**, and find your way with **Campus Maps.**

Download the Sydney Uni app

Visit myuni.sydney.edu.au or download the Sydney Uni app.

Accessing assistance at our in-person hubs

Need in person assistance on campus? Visit our Student Services Hubs, where we can help you with:

- Completing enrolment
- Understanding your timetable
- Course & Degree planning
- Candidature variations
- ICT support
- Peer and learning support
- General advice to thrive at The University of Sydney

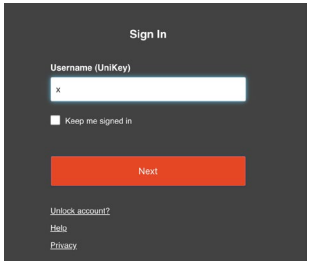
Find a hub

Student Services Hubs are open **from 27 July to 31 August, Monday to Friday, 9 am to 5 pm** across multiple campus locations.

You can collect your Student ID card from our Cards Hub, **after the date in your 'Ready to Collect' email** O. The Cards Hub will temporarily relocate to the foyer of the Wilkinson Building from 20 July to 31 August (census date) to support Semester 2 commencement**.** Cards cannot be printed prior to this date. For instructions on initiating a card request, visit the Student Cards website.

Managing access to our University systems

Multifactor Authentication (MFA) is an additional security step to verify your identity at the University.



Step 1
Sign in with your username (UniKey and password)

Step 2
Verify your identity using your smartphone

For more information
Scan the QR code

Remember:

- Set up a secondary email, phone number and security question in your Okta profile so that you can reset your Unikey.
- Always have your Okta enabled device with you and do not uninstall the Okta verify app.
- If you change your mobile device, install Okta Verify on your new device before removing it from your old one.

School of Physics Equity, Diversity and Culture Committee

In the School of Physics, we value diversity, a welcoming culture and providing equal opportunities.

The School of Physics Equity, Diversity and Culture Committee runs initiatives in the School that improve workplace culture and inclusion; and advises the School Leadership team on Culture or EDI issues in the School;

Get involved!

1) Committee email: physics.edc.committee@sydney.edu.au or chair email: theresa.fruth@sydney.edu.au.
   Feel free to email us about any issues you would like to raise.

2) Mail list for School members interested in EDC issues: physic.edc.community@sydney.edu.au
   Sign up: School of Physics - Equity, Diversity and Culture Community Email Sign-Up – Fill out form

3) You have your own representative on the committee, currently Shaun Parasher
   Contact through the PhySoc usydphysoc@gmail.com

Academic Integrity: What you need to know

THE UNIVERSITY OF SYDNEY

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Academic integrity is…

"a commitment to five fundamental values: ***honesty, trust, responsibility, fairness, respect*** *… plus the* ***courage*** *to act on them even in the face of adversity..."*

The Fundamental Values of Academic Integrity, International Center for Academic Integrity (Third Edition)

Acting with *academic integrity* means:

- Following ethical principles in all dealings at university
- Following the rules set for assignments and exams
- Doing your own work and never 'outsourcing'
- Acknowledging *any* contributions which are not your original thoughts, ideas or words
- When working in groups, doing so legitimately – i.e., not colluding

Academic integrity goes to the heart of what a university's role is: educating students, advancing research, contributing to the world's knowledge, and qualifying people to do particular jobs. Without integrity, a university's activities lose meaning and value.

Types of academic integrity breaches

Plagiarism

Presenting work that is not your own without acknowledging the original source.

Collusion

Engaging in illegitimate cooperation with one or more students in the completion of assessable work.

Fabricating information

Making up information, such as experimental or interview data. It can also include inventing sources by citing publications that are incorrect or that don't exist.

Recycling

Re-submitting work that has already been assessed without your teacher's permission and for which you have already been given feedback.

Contract cheating

This involves having someone else complete all or part of your work for you. It includes

- buying a completed assignment
- having someone else write part or all of an assignment
- using model answers from a tutoring company
- engaging in the buying, selling, and swapping of assignments.

Misuse of AI

Using generative AI tools when they have been prohibited for assessment, or, where AI use is allowed, failing to acknowledge what tools you have used and how.

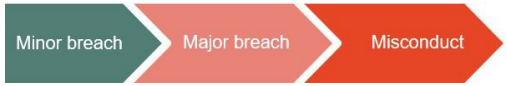
Exam cheating

Breaching exam conditions, including using unpermitted materials or devices, communicating with others or allowing someone else to take the exam on your behalf.

sydney.edu.au/students/academic-integrity

The University's approach

The University **investigates** any student suspected of breaching the Academic Integrity Policy. Teachers monitor submissions for potential integrity issues and report any suspected breaches.



Substantiated breaches are classified in three categories:
minor breach, **major breach**, or **misconduct**.
The investigation process and penalties differ depending on the type of breach.

Potential penalties for breaches include:

- Marks deducted for an assignment or exam
- Requirement to do a 'development' activity
- Failure of the assessment task or the unit of study
- Suspension or exclusion (for more serious breaches)
- If you are an international student, you could even lose your student visa.

NOTE: The personal impact of investigations and adverse findings can go beyond the academic penalties.

Complete the *Academic Honesty Education Module* (AHEM)

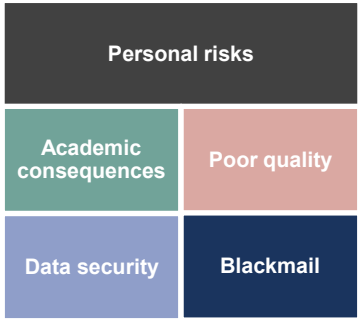
The Academic Honesty Education Module ('AHEM') is compulsory for all new students, while other students are strongly encouraged to complete it.

The AHEM provides essential information about academic integrity, and it typically takes about 1 hour to complete. It covers the key principles of academic integrity, the risks of engaging in breaches, and provides guidance on where to seek support.

New students must complete the AHEM before the census date (1 September 2025). The Module will appear in your Canvas dashboard at the start of semester.

Failure to complete the module could result in your academic results being withheld.

Be aware of contract cheating providers: *Know the risks*



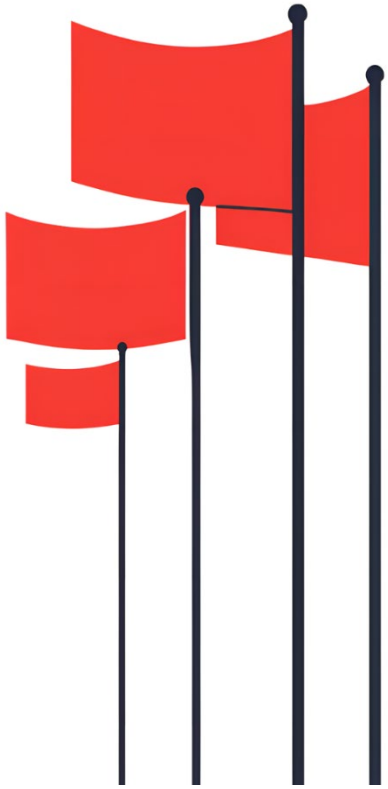
Contract cheating providers target students online and in person, sometimes even on campus such as at Welcome events. They often call themselves 'tutoring' and may offer other services alongside academic services such as housing advice, course advice, and appeals assistance.

Advertising or providing contract cheating services is against the law. Penalties can include fines of up to $100,000 or imprisonment.

There are many risks….

- Contract cheating is considered **academic misconduct** and results in severe penalties – including failing the assignment or unit, suspension or exclusion.
- The quality of work is often **poor, badly written or wrong**. In many cases it doesn't even get the student the desired grade.
- When you engage with these companies, you not only put yourself at risk of being caught, but also expose yourself to a range of other risks including **being blackmailed** by the company, harassed, and having your personal data/privacy compromised.

Signs you might be dealing with a *contract cheating service*



- You're approached or solicited to use the service in person, or online
- You're asked to pay for the service
- You're asked to share your Unikey or other personal details
- Advertisements for the service mention a specific USYD unit of study
- The service provides a 'guarantee' that you won't get caught
- The service is not acting in an open, transparent way
- You're added to a private group chat by the service
- You're given step-by-step instructions for how to complete an assessment
- The service is aggressive or threatening
- You feel that by using the service you are getting an unfair advantage

To report a service or if you find yourself in a difficult situation, contact the Office of Educational Integrity

QR code: How to spot a contract cheating service

Using generative AI

Generative AI tools, such as **Microsoft Copilot Chat**, can create digital content, including text, images, and video. Different guidelines for use of generative AI apply to learning and assessment:

Generative AI for general learning

- You can use generative AI to help you learn. For example, it can explain things, apply knowledge, plan your study and make practice questions. Use our student co-designed AI resource so you are aware of its strengths and limitations.

Generative AI for assessment

- Check your unit outline for whether you can use generative AI tools in your assessment:
    - For in person supervised tasks like exams – generally you will not be allowed to use AI unless your coordinator expressly permits it.
    - For other tasks, generally you will be able to use it, but you must appropriately acknowledge all use.
- When using generative AI tools, don't provide content generated by other students, others' intellectual property (unless with permission), or personal or health information. Your unit coordinator can provide guidance on this. Failing to acknowledge AI use is an academic integrity breach.

AI resource made by students:
https://bit.ly/students-ai

Which generative AI tools are best to use?

The University provides **free** access to Microsoft Copilot Chat to accelerate your learning and help you in those assessments where use of generative AI is allowed.



Why use Microsoft Copilot Chat?

- It is a University-supported, industry standard tool powered by up-to-date AI models
- It keeps your data secure and protects your privacy
- Log in with your UniKey using Okta so that you are using Copilot in protected mode, which introduces necessary guardrails to protect you within the system



Why not use other tools like ChatGPT and DeepSeek?

- Watch out for your privacy and the security of your personal and other data if using a free or emerging tool like ChatGPT and DeepSeek

Powerful and secure
https://copilot.microsoft.com/

File-sharing

File-sharing sites allow you to buy, trade or sell academic material. Using these sites puts you at risk of academic integrity breaches.

Do not upload:

- completed assessments (which can be used by other students to cheat) – this is a breach of the Academic Integrity Policy.
- University course or teaching materials, including assignment questions – this is a breach of University copyright and the Academic Integrity Policy.



Legitimate support

Contact your unit of study coordinator or teaching staff for guidance and support.

Learning Hub (Wentworth Building and online)

- Learning Hub – Academic Language
- Learning Hub – Mathematics
- Leaning Hub Canvas site and SCANA test
- Peer programs – “Speak + Connect” and “Speak Up”

Library

- Referencing guide
- Meet with a Peer Learning Advisor
- PeerConnect (online chat)
- Exam Ready sessions

Other resources

- AI in Education (Canvas site on generative AI)
- Study Skills page of the student website
- The Write Site (Canvas site)
- Power up your learning (Canvas site)
- Academic Progression

Advice about integrity investigation

- SRC Caseworkers (undergraduates)
- SUPRA Caseworkers (postgraduates)

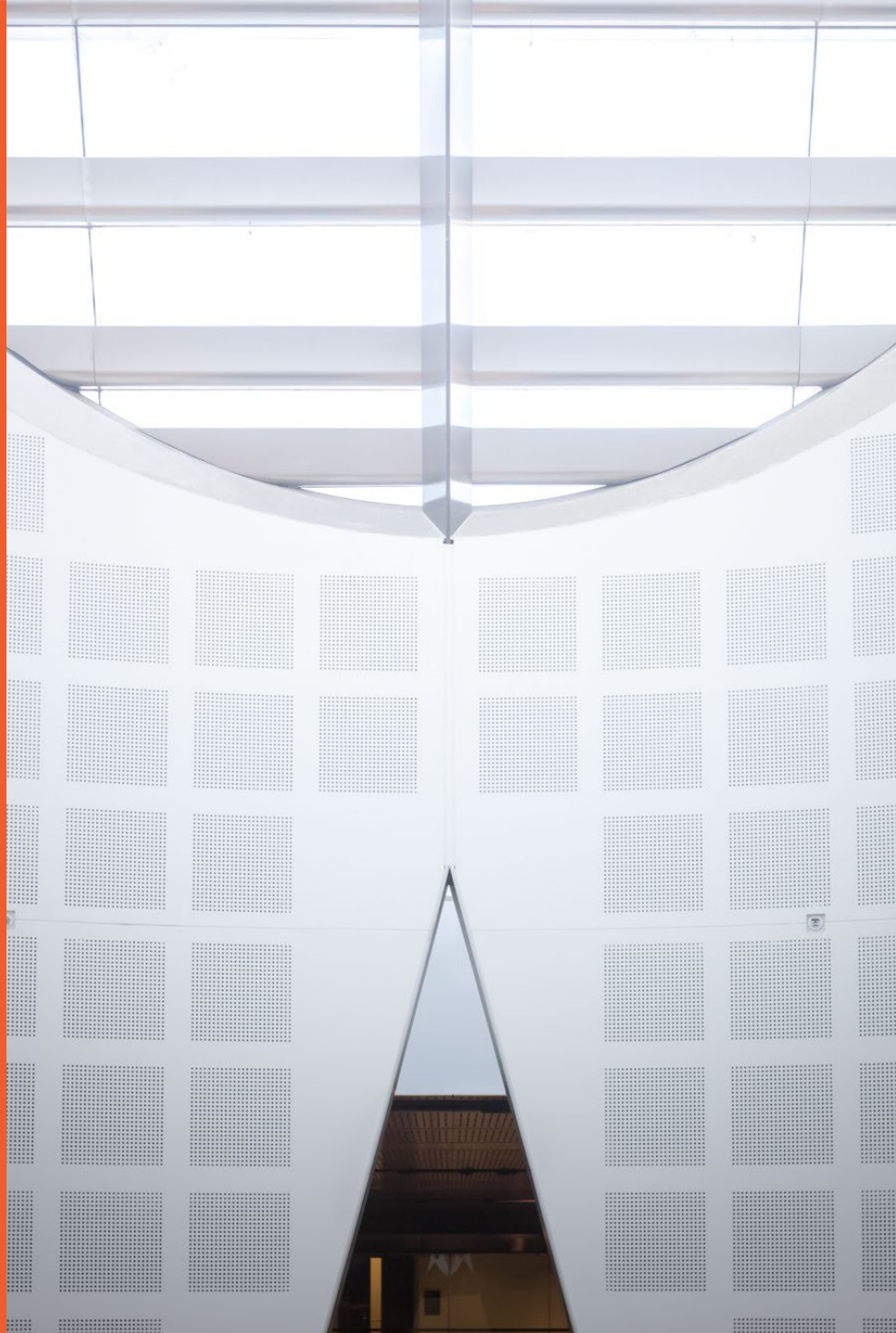
Questions?



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PHYS5033



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PHYS5033 – What will I learn in this unit?

- What supply chains are, and why we want to assess them
- What input-output tables are, and how they are structured
- How input-output tables are generated, and where we can find them
- What footprints are, and how to calculate them using input-output tables
- What is meant by "impacts embodied in supply chains"
- What is meant by triple bottom line accounting
- Hands-on practice in Excel and MATLAB
- Guest lecturers from academia and industry

Where can I use these skills?

- Industry – eg sustainability consultant
- Councils
- Academia
- Policy-making

This unit of study will provide students with practical skills for carrying out environmental footprinting calculations – for individuals, companies, organisations or nations. In particular, this unit will provide a comprehensive introduction to input-output analysis for identifying impacts embodied in regional, national and global supply chains. This unit focuses on contemporary environmental applications such as emissions, energy-use, water, land, loss of animal & plant species; and also social applications such as employment, poverty and child labour. The unit first explores national and global economic and environmental accounting systems and their relationships to organisational accounting. Second, it presents cutting-edge techniques enabling the global analysis of environmental and social impacts of international trade. Third, it offers hands-on practical activities for mastering the input-output techniques conceived by Nobel Prize Laureate Wassily Leontief, and provides a step-by-step recipe for undertaking boundary-free environmental and social footprinting for sectors and organisations. Students will walk away from this unit equipped with useful skills needed to calculate footprints, and prepare sustainability reports for any organisation, city, region, or nation, using organisational data, economic input-output tables and environmental accounts.

PHYS5033 organisation

Weekly lectures and tutorials

Weeks 1-8, 10-13: Mondays, 12noon - 3pm in Old Geology Lecture Theatre, Edgeworth David Building

Week 9: Thursday 8 October, 1 - 4pm in Veterinary Science Conference Centre. VSCC Lecture Theatre 208 (Webster)

Lectures and tutorials start in the first week of semester.

This unit will largely include hands-on activities for learning the techniques for footprint assessments. Please make sure you have Excel ready to go on your laptops, as we will be working on Excel-related activities, starting Week 1.

All lectures will be recorded and posted on Canvas under the "Recorded Lectures" tab

PHYS5033 organisation – Questions

- Questions are encouraged
- The only dumb question is the one that doesn't get asked!

PHYS5033 organisation – assessment tasks

Weighting	Description
40%	**Sustainability report** This task requires students to prepare a sustainability report for a Council, providing quantitative figures for the calculation of their personal carbon footprint, based on techniques discussed in the lectures. Furthermore, students are required to provide a thorough discussion and a plan of action for reducing the footprint. Submissions are due on Turnitin by 11:59PM, 16 November, 2026. **See detailed guidelines below.**
50%	**Homework Submission 1 and 2** Consistent completion of weekly homework tasks: Homework Submission 1 Due Date: 9 September, 2026, 11:59PM (Tasks 1-5; from weeks 1-5) Homework Submission 2 Due Date: 28 October, 2026, 11:59PM (Tasks 6-10; from weeks 6-10) Students need to submit an Excel and a Word/PDF file for each Homework Submission. **See detailed guidelines below.**
10%	**Multiple choice tests** There will be a multiple choice test in Weeks 4, 6, 8, 10 and 12. –QUIZ 1 - WEEK 4 - Questions will be based on material covered in Lectures 1 & 2 (open: 3:00pm 24 August - 3:00pm 25 August) –QUIZ 2 - WEEK 6 - Questions will be based on material covered in Lectures 3 & 4 (open: 3:00pm 7 September - 3:00pm 8 September) –QUIZ 3 - WEEK 8 - Questions will be based on material covered in Lectures 5 & 6 (open: 3:00pm 21 September - 3:00pm 22 September) –QUIZ 4 - WEEK 10 - Questions will be based on material covered in Lectures 7 & 8 (open: 3:00pm 12 October - 3:00pm 13 October) –QUIZ 5 - WEEK 12 - Questions will be based on material covered in Lectures 9 & 11 (open: 3:00pm 26 October - 3:00pm 27 October) Students will be able to login anytime between the opening hours of the quiz to complete the questions. There will be 10 questions, 15 minutes to complete the quiz. Students will have two attempts to complete the questions. Best mark out of the two attempts will be considered. **The quiz will be open-book**. Two sample quizzes have been opened for you to see how the quiz will run. Students are responsible for ensuring that they login between the opening hours of the quiz for completing the questions. Unfortunately, it is not possible to run a replacement quiz. If students miss a quiz due to unforeseen circumstances, they can apply for special consideration. If successful, their mark will be estimated based on their performance in other quizzes. An extension of quiz time is not possible.

PHYS5033 organisation – assessment tasks

Homework Tasks

Each week, you will build your skills in undertaking footprint assessments through a step-by-step learning process. Developing confidence and proficiency comes with regular practice, so these homework tasks are designed to help you apply the concepts covered each week by working on your own input–output table of the specified dimension.

Please complete the weekly homework tasks listed under the **Modules** section, using the concepts and methods taught during that week's learning activities.

Please take note of the following requirements for homework submission:

Homework Submission 1

Worth: 25% of final mark
Due: 9 September, 2026, 11:59PM

You must submit **two separate files**:

Excel File 1

Submit under **Assignments** (**Homework Submission 1: Excel File**)

Your Excel workbook should:

- Include completed homework tasks for **Weeks 1–5**.
- Use **a separate worksheet (tab) for each week**.
- Display **all formulae** used in calculations (do not paste values only).
- Clearly label **all data, variables, and units**.

Important:
Your submission **must include the required number of sectors specified in each week's homework task. If your submission does not include the required number of sectors, a penalty of 50% of the maximum mark available for that week's homework will be applied.**

Homework Submission 2

Worth: 25% of final mark
Due: 28 October, 2026, 11:59PM

You must submit **two separate files**:

Explanation File (Word or PDF) 2

Submit under **Assignments** (**Homework Submission 1: Word/PDF File**)

Prepare a Word document containing your answers to the **Explanation** questions for **Weeks 1–5**. You can either submit a Word file or a PDF file.

The Explanation questions are designed to assess your understanding of the methodology underlying input–output calculations.

Important:

- Explanation answers included in the Excel file **will not be marked**.
- Failure to submit the Explanation file will result in **zero marks for the Explanation component** of Homework Assignment 1.
- Explanation **answers that include terminology that is unrelated to the content taught in this unit may have marks deducted**.

PHYS5033 organisation – assessment tasks

Tips for the Homework tasks:

- Make sure that you read the homework instructions thoroughly for each week – if you don't complete one of the steps expected, you won't get any marks for it.
- If there is a question posed in one of the homework steps, make sure you provide an explicit answer for that
- Always orkings are included
- You will xamples of how the If you just replicat
- You sho ework each week us There is no need to s providing irreleva st be cited and referenced appropriately. You are strongly advised NOT to rely on AI tools to source your answers for these questions since they usually provide incorrect information.
- Make sure that your Excel submission is easy for the marker to read and find your results. This includes using appropriate font sizes, using appropriate row heights and column widths, and including text to note which step the results you're presenting relate to.

Always refer to the information on Canvas to ensure you complete the tasks as expected

PHYS5033 organisation – assessment tasks

Sustainability report

Worth: 40% of final mark

Word limit: 3000 words

Due: 16 November, 2026

Download marking criteria

Note

Scenario

You have been asked by your local council to prepare a sustainability report outlining the methodology for calculating your carbon footprint as a resident of that council. Furthermore, the sustainability manager of the council has asked you to provide concrete examples and recommendations on ways for reducing your carbon footprint, and that of other residents of the council.

Note: This is an exercise to get you to engage with the technique of input-output analysis for calculation of individual footprints, and to get you thinking about ways of (and the need for!) reducing greenhouse gas emissions. If you do not wish to reveal your real spending pattern (see Part A, point 1 below), then you are free to use dummy data for the purpose of this assignment. It is important, however, to make sure that the dummy data that you use is realistic.

Your task

You have been provided with the following data files for Year 2018:

Intermediate demand matrix (T) – 345 x 345 matrix (data on 345 sectors of the Australian economy. See 'Labels file' for more information). The units for these data are Australian million dollars. For example, a value of 2 in a cell means 2 million Australian dollars.

Total output vector (x) – 345 x 1 vector (data on total output for 345 sectors of the Australian economy). The units for these data are Australian million dollars.

Satellite account (Q) – 1 x 345 vector (data on carbon dioxide equivalent emissions for 345 sectors of the Australian economy). The units for these data are kilotonnes.

These files are stored in two directories: MATLAB files and Excel files. Data in both directories are the same. You can complete this assignment using either the MATLAB programming software (in that case use the MATLAB files) or MS Excel (use Excel files).

PHYS5033 organisation – assessment tasks

Sustainability report

Worth: 40% of final mark

Word limit: 3000 words

Due: 16 November, 2026

You are required to undertake the following steps:

PART A: QUANTITATIVE CALCULATIONS

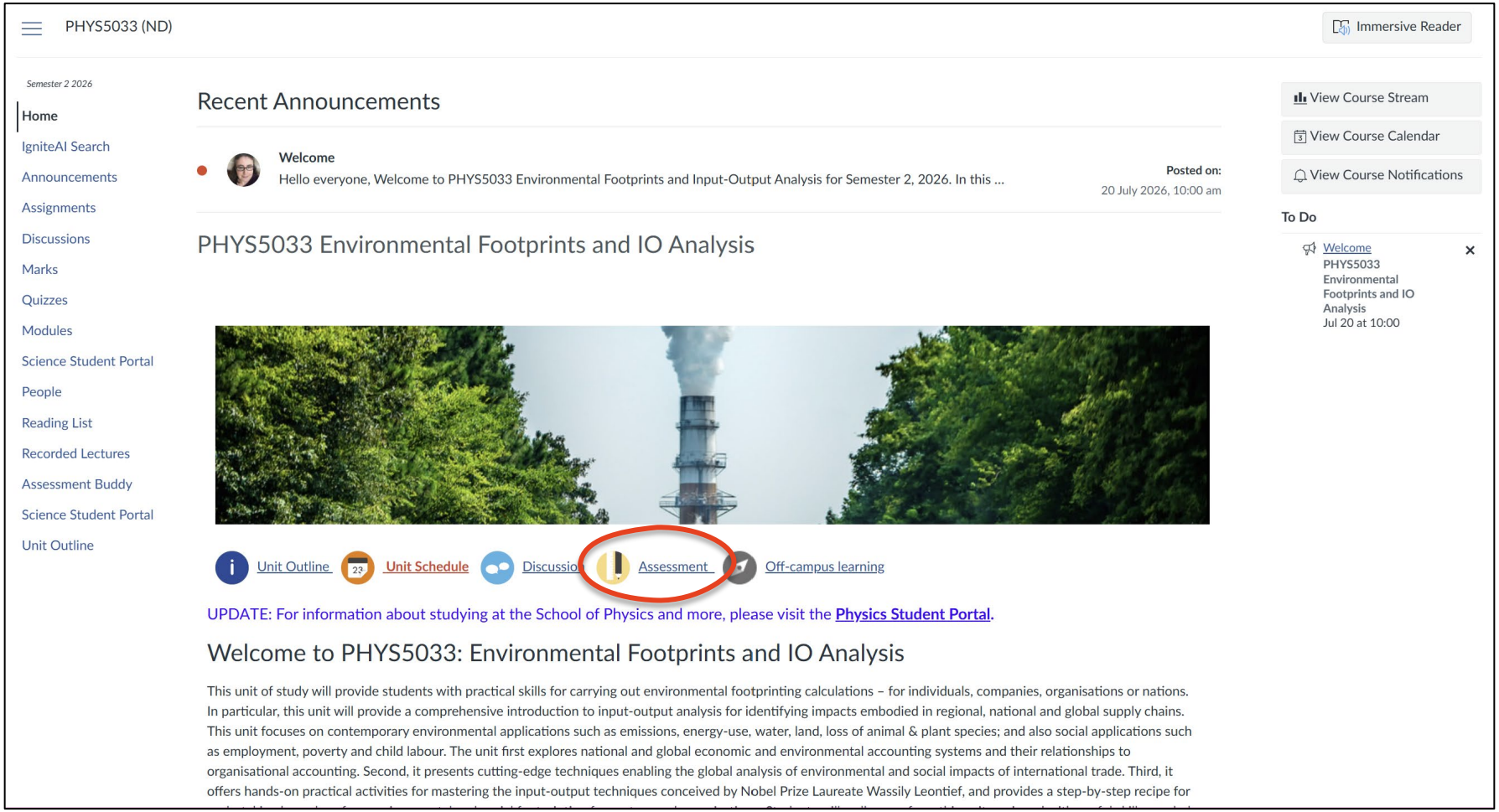
- *Collection of Expenditure Data*: What do you spend money on? Collection of yearly expenditure data on various consumables (food products, electricity, fuel, transportation, clothing, furniture, services etc.).
- *Construction of final demand shopping basket vector*: Map your raw expenditure data to the 345 input-output categories.
- *Calculation of CO_2e footprint*: Use the consumption-based footprint equation to calculate your carbon footprint.
- *Comparison of per-capita footprints*: Compare your carbon footprint figure with: a) per-capita footprint of an average Australian; and b) per-capita footprints in other countries. Present your results in the form of figures/tables. Example - see here: https://ourworldindata.org/co2-and-other-greenhouse-gas-emissions#co2-embedded-in-trade

PART B: REPORT WRITING

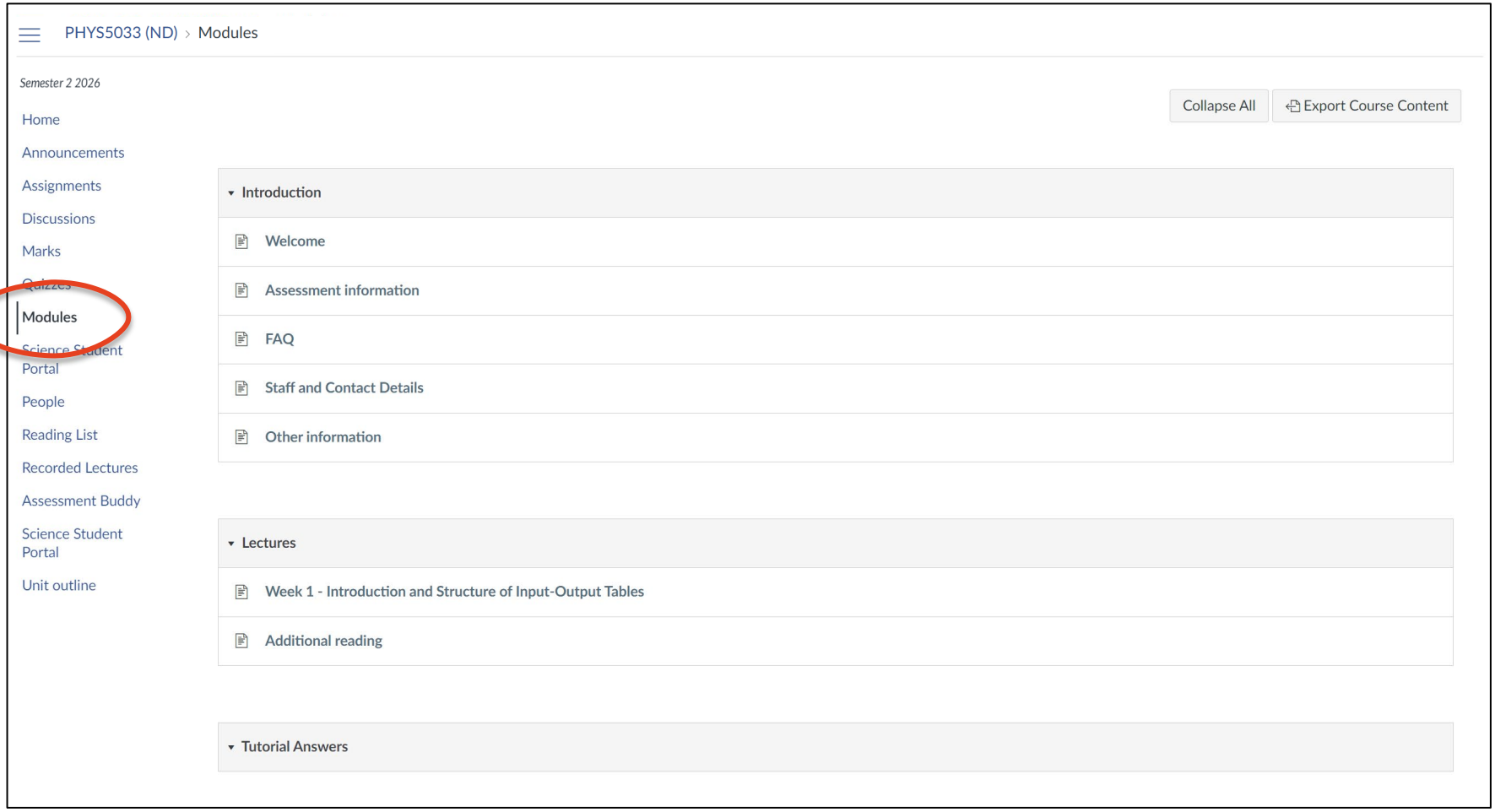
Put together a report for your local council showcasing the results of the assessment, and outlining strategies for reducing carbon emissions. Your report should include the following information:

- *Introduction*: The need for a reduction in greenhouse gas emissions. Provide a national and a global perspective;
- *Introduction*: The case for the use of quantitative methods, such as input-output analysis, for analyzing footprints;
- *Methodology*: Methodology underlying input-output analysis;

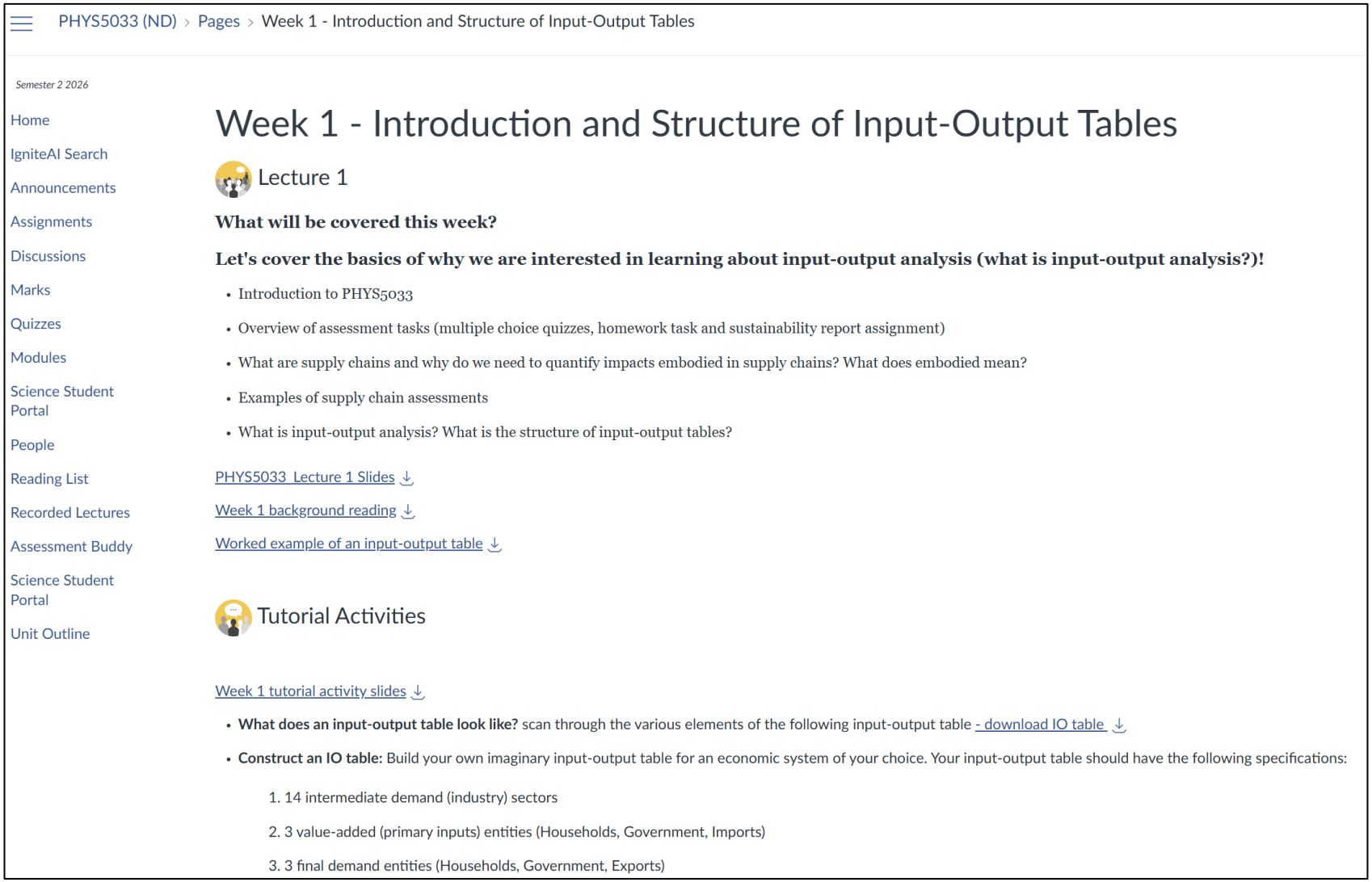
Canvas Navigation



Canvas Navigation



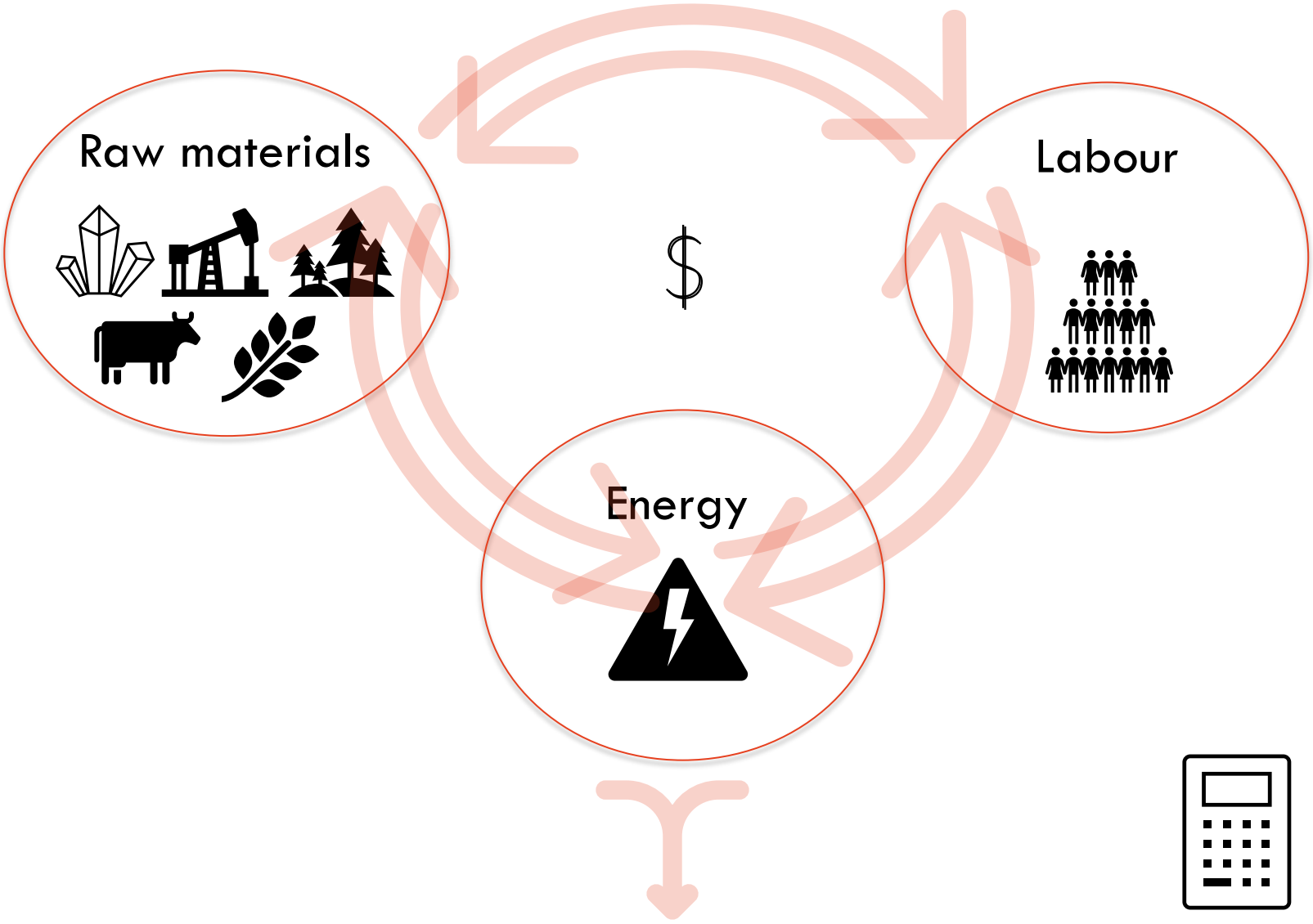
Canvas Navigation

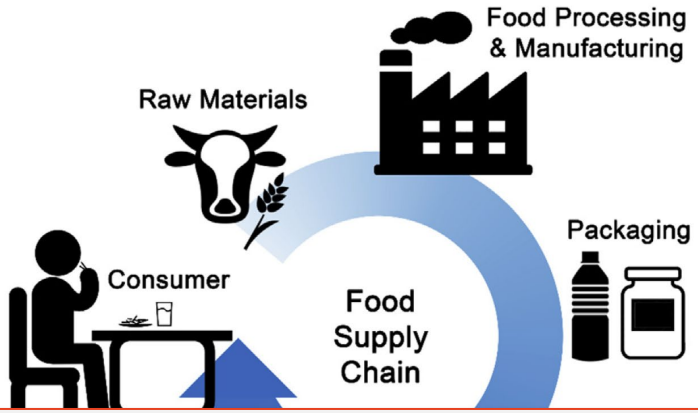


Why assess supply-chains?



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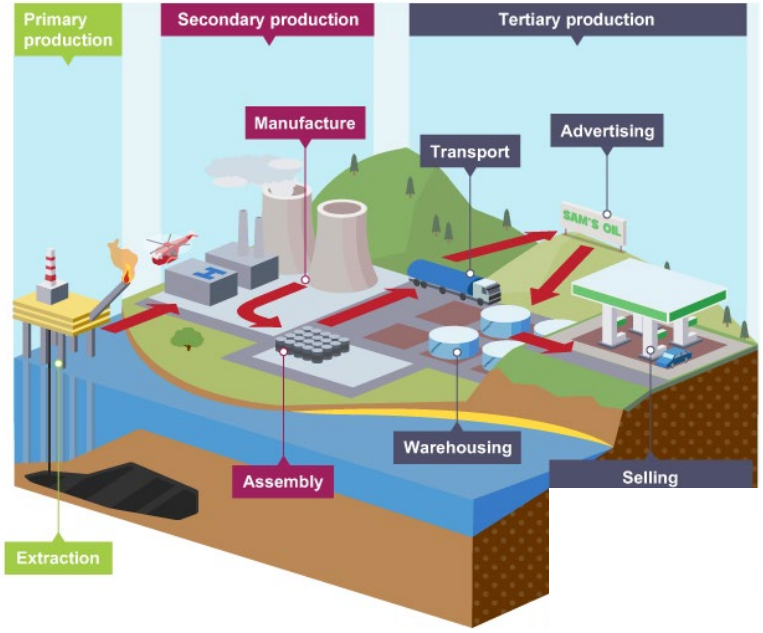

Raw materials
$
Labour
Energy
Consumption
The economy



What are some examples of supply chains?

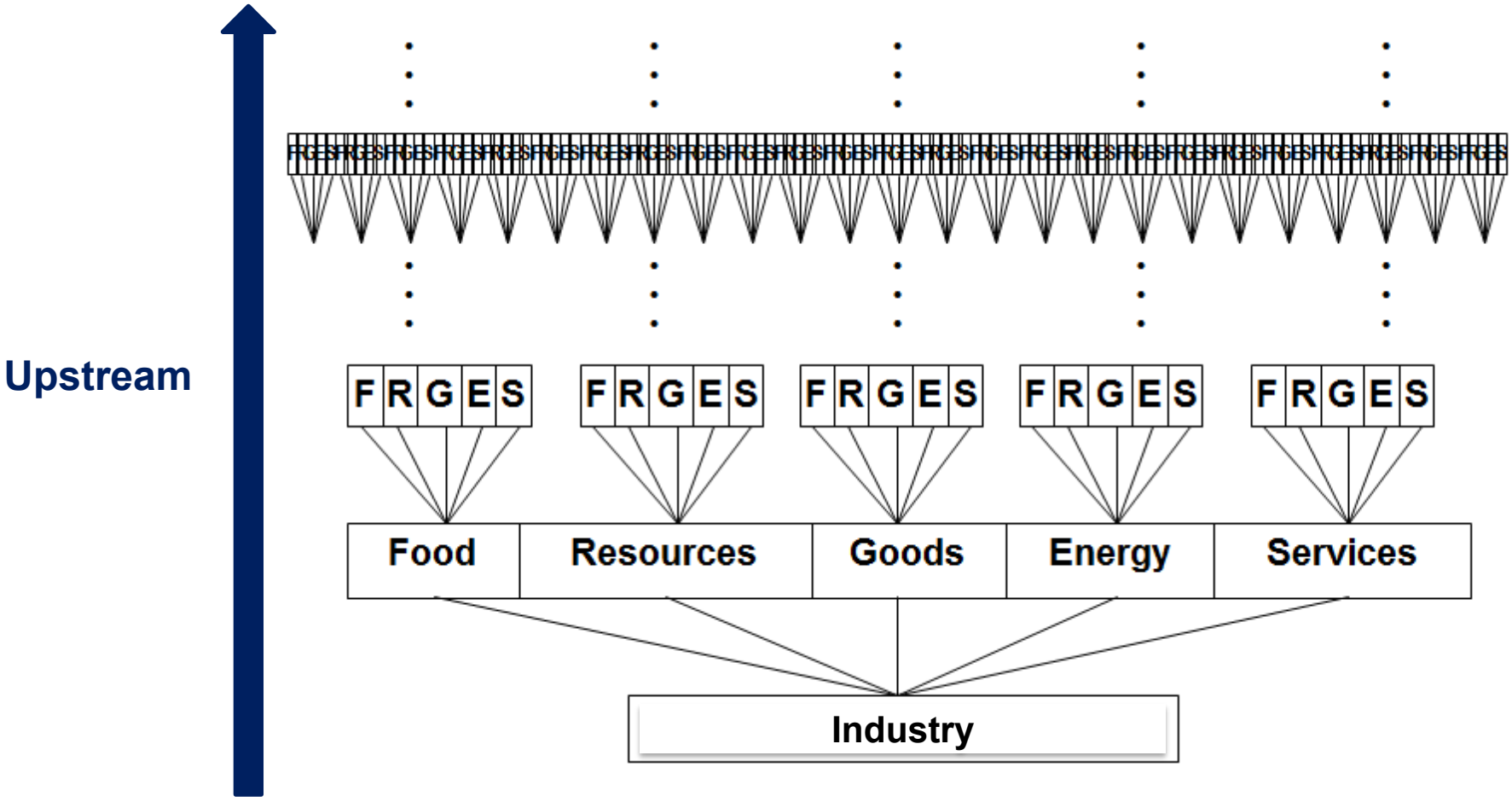
Tracking the flow of money - economic sectors

- Primary sectors: extraction of raw materials (mining, fishing, agriculture)
- Secondary sectors: production of intermediate and/or finished goods (cars, food, clothes, shoes, electricity)
- Tertiary sectors: service sectors (banking, IT services, consultancy)

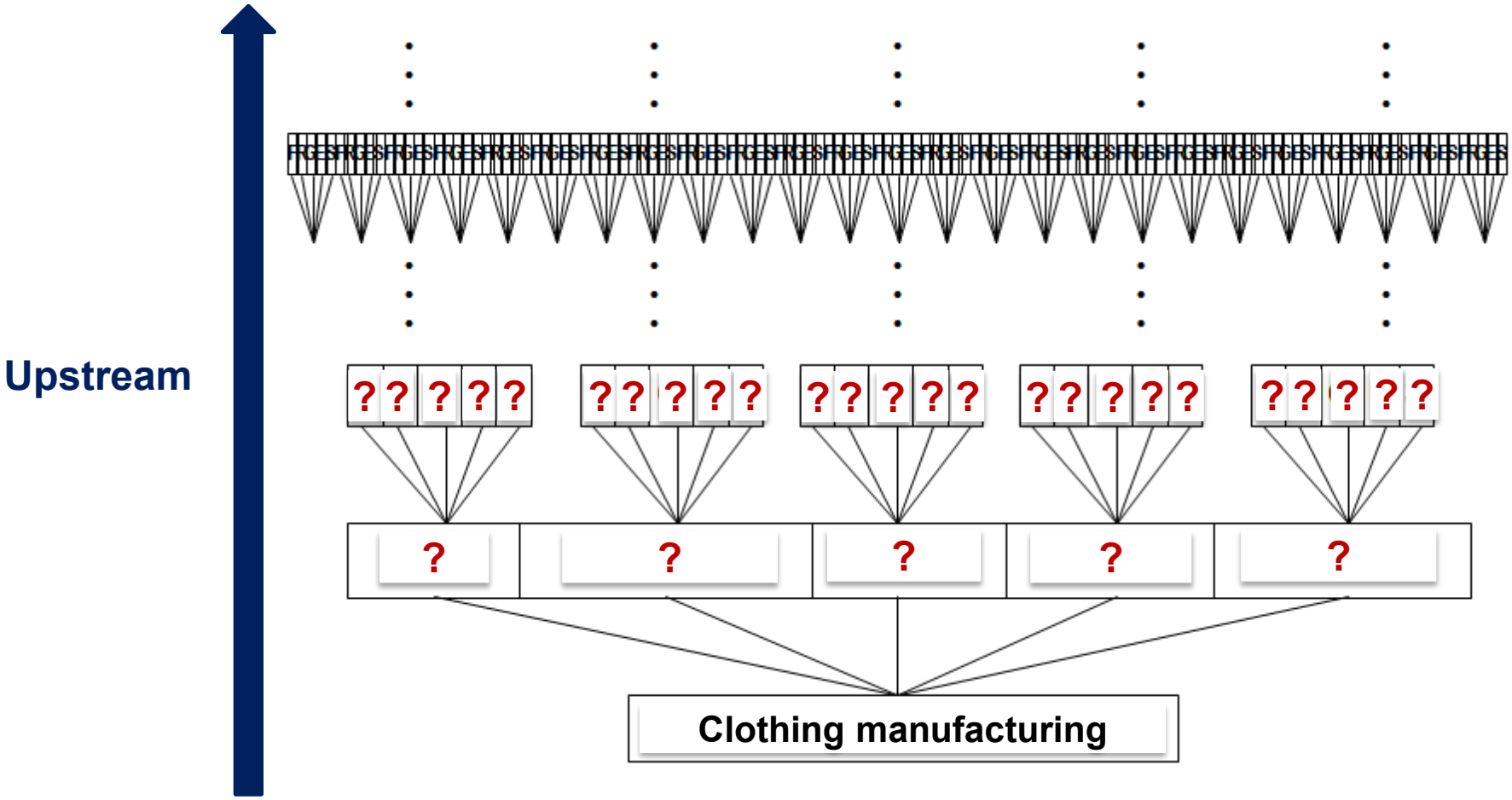


https://www.bbc.co.uk/bitesize/guides/zrvb9j6/revision/3

Supply chain analysis



Activity – unravelling upstream supply chains



Supply Chain Sustainability assessments – a snapshot

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Supply chain assessment – biodiversity threats

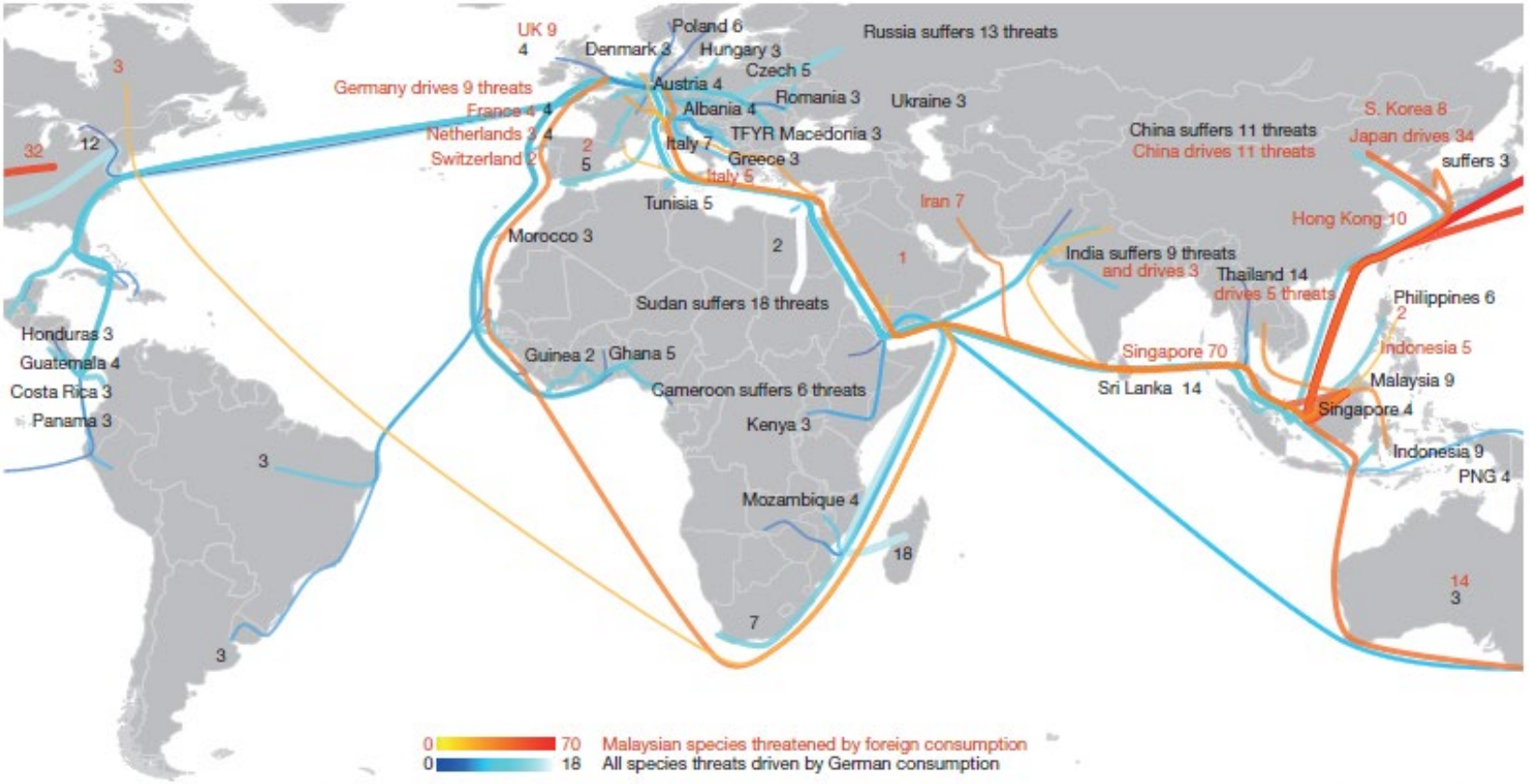


Figure 3 | Flow map of threats to species. Flow map of threats to species caused by exports from Malaysia (reds) and imports into Germany (blues). Note that the lines directly link the producing countries, where threats are recorded, and final consumer countries. Supply-chain links in intermediate countries are accounted for but not explicitly visualized. An interactive version is available at http://www.worldmrio.com/biodivmap/.

Lenzen M, Moran D, Kanemoto K et al. International trade drives biodiversity threats in developing nations. Nature, 2012; 486 (7401):109-12.

Supply chain assessment – impacts of COVID-19

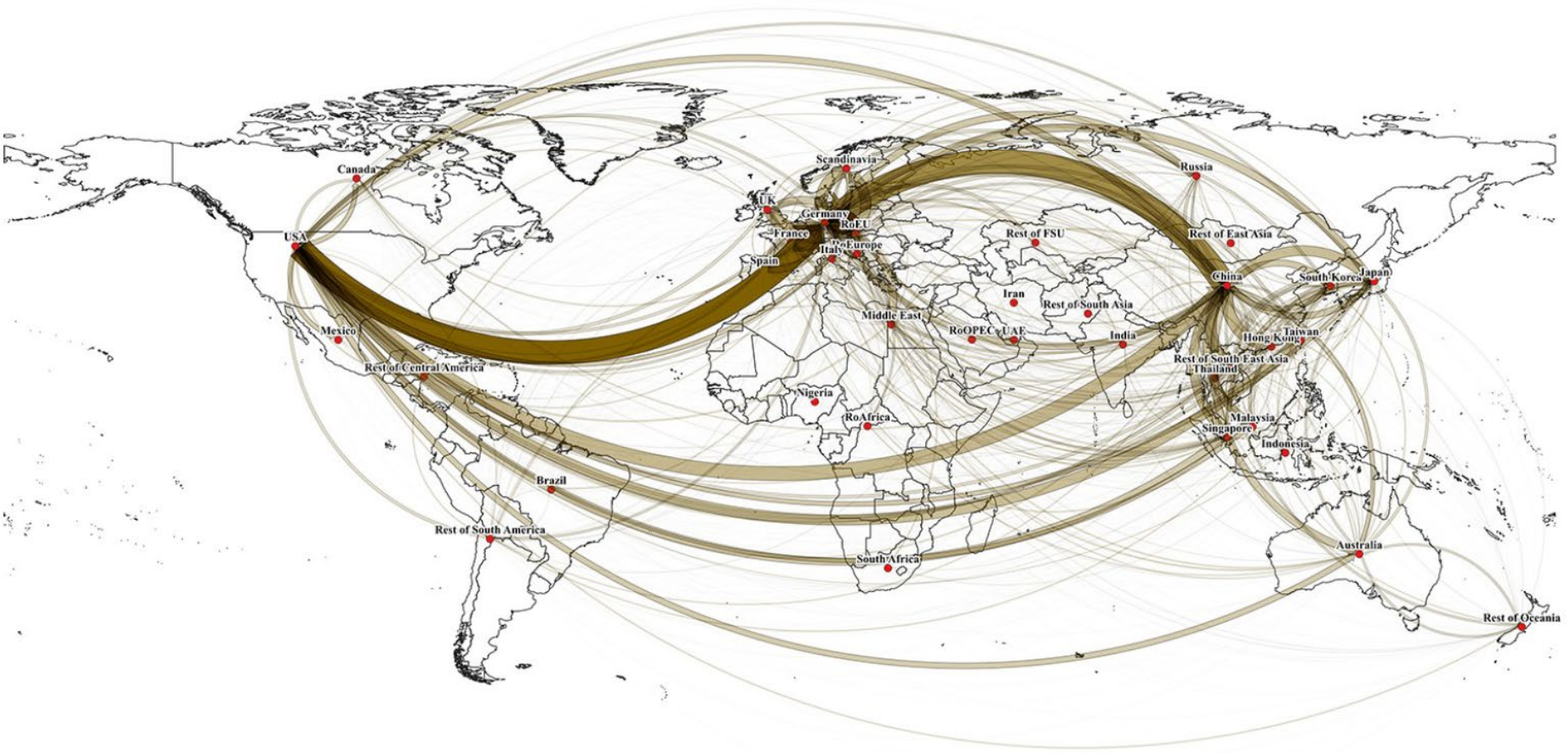
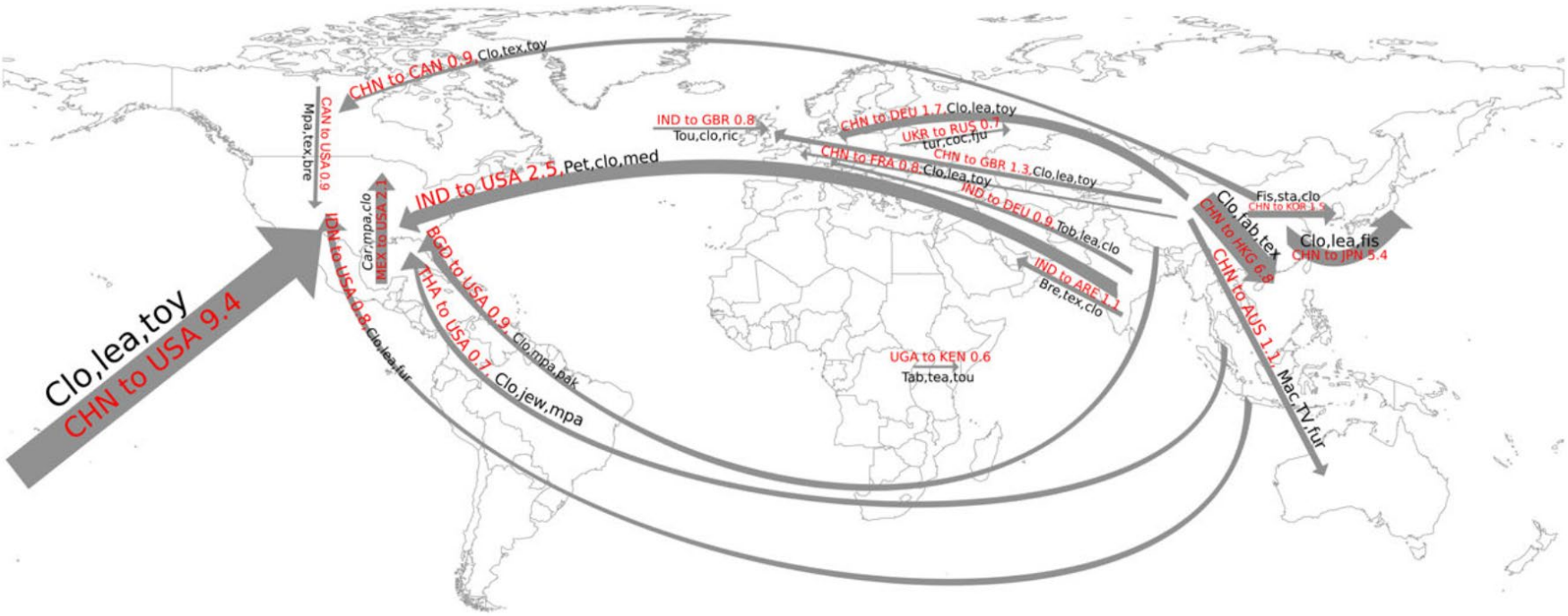


Fig 3. Wage and salary income losses as a consequence of trade volume reductions in international supply chains due to the global COVID-19 effects. Lines connect ultimate origins and destinations of supply chains, both direct and multi-node. Line thickness represents trade volume lost.

Lenzen, M., Li, M., Malik, A., Pomponi, F., Sun, Y.-Y., Wiedmann, T., et al.(2020), "Global socio-economic losses and environmental gains from the Coronavirus pandemic", *PLoS ONE*, Vol. 15 No. 7, pp. 1-13.

Corruption embodied in production of goods



Xiao, Y., M. Lenzen, C. Benoît-Norris, G. A. Norris, J. Murray, and A. Malik. 2018. The corruption footprints of nations. *Journal of Industrial Ecology* 22(1): 68-78.

Sustainability assessment



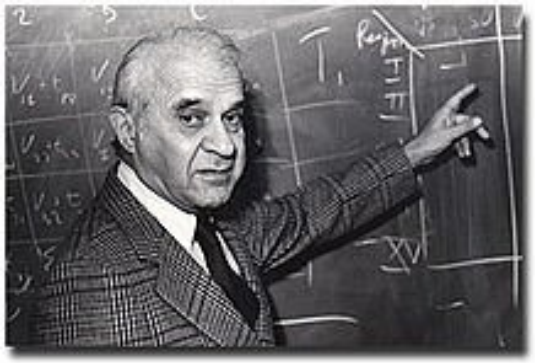
Introduction to input-output tables



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Input-output analysis

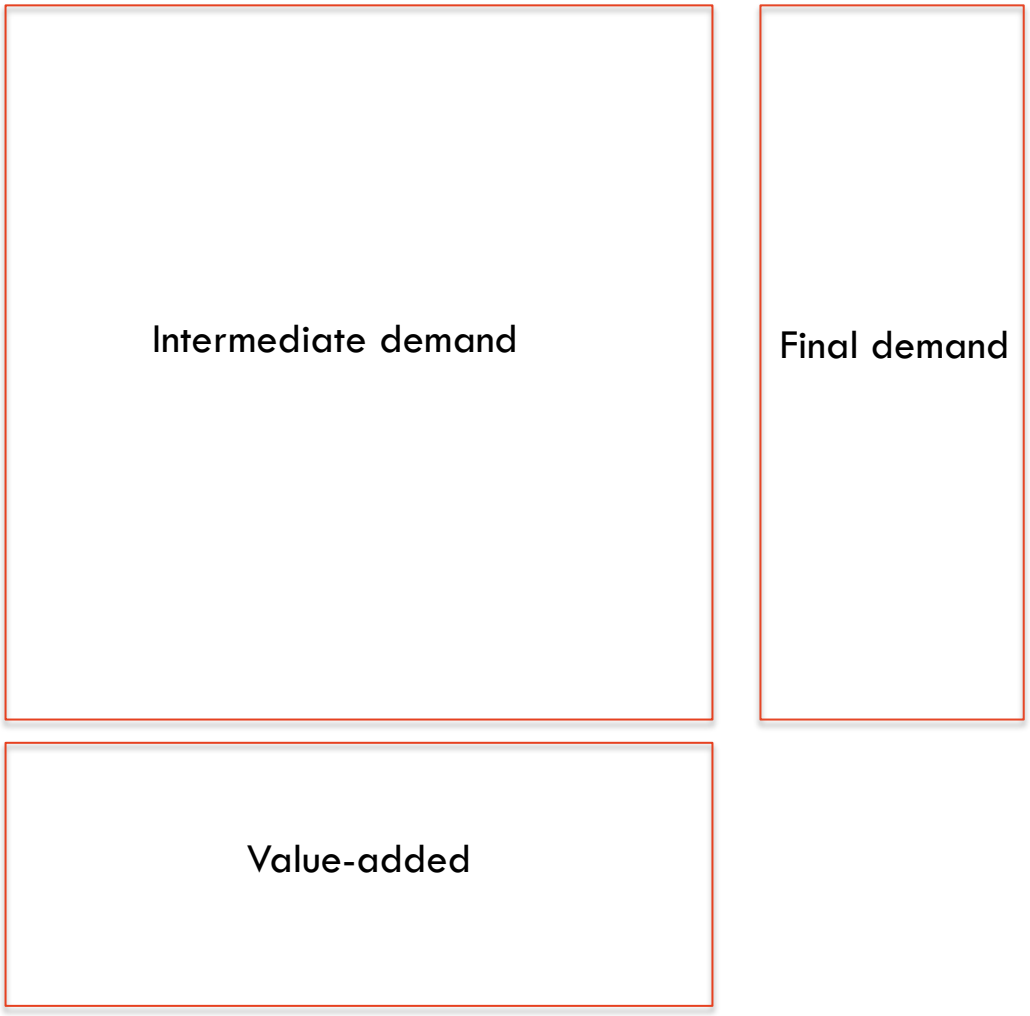
- Relies on national input-output tables
- Input-output (IO) tables document the flow of money between sectors in an economy
- The methodology has been around since the 1930s



Wassily Leontief – Nobel Prize 1973

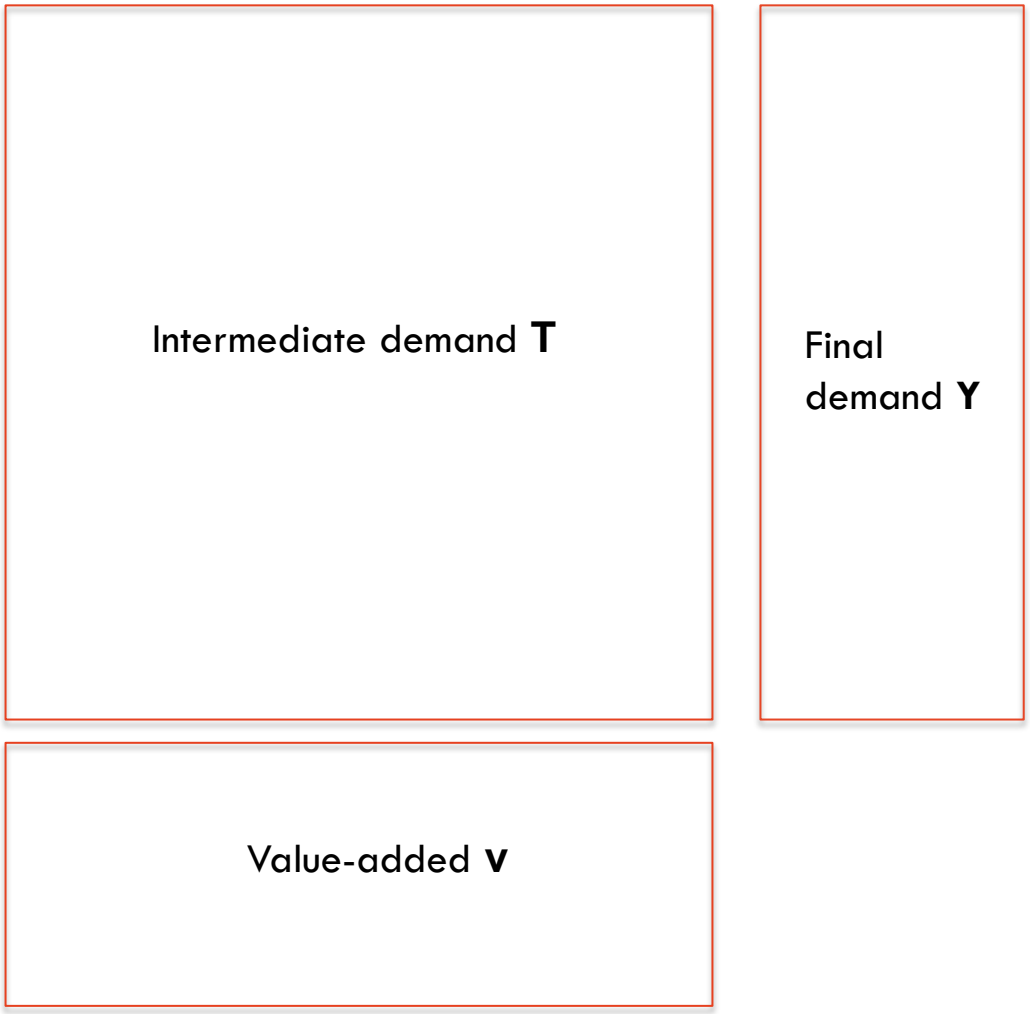


3 matrices



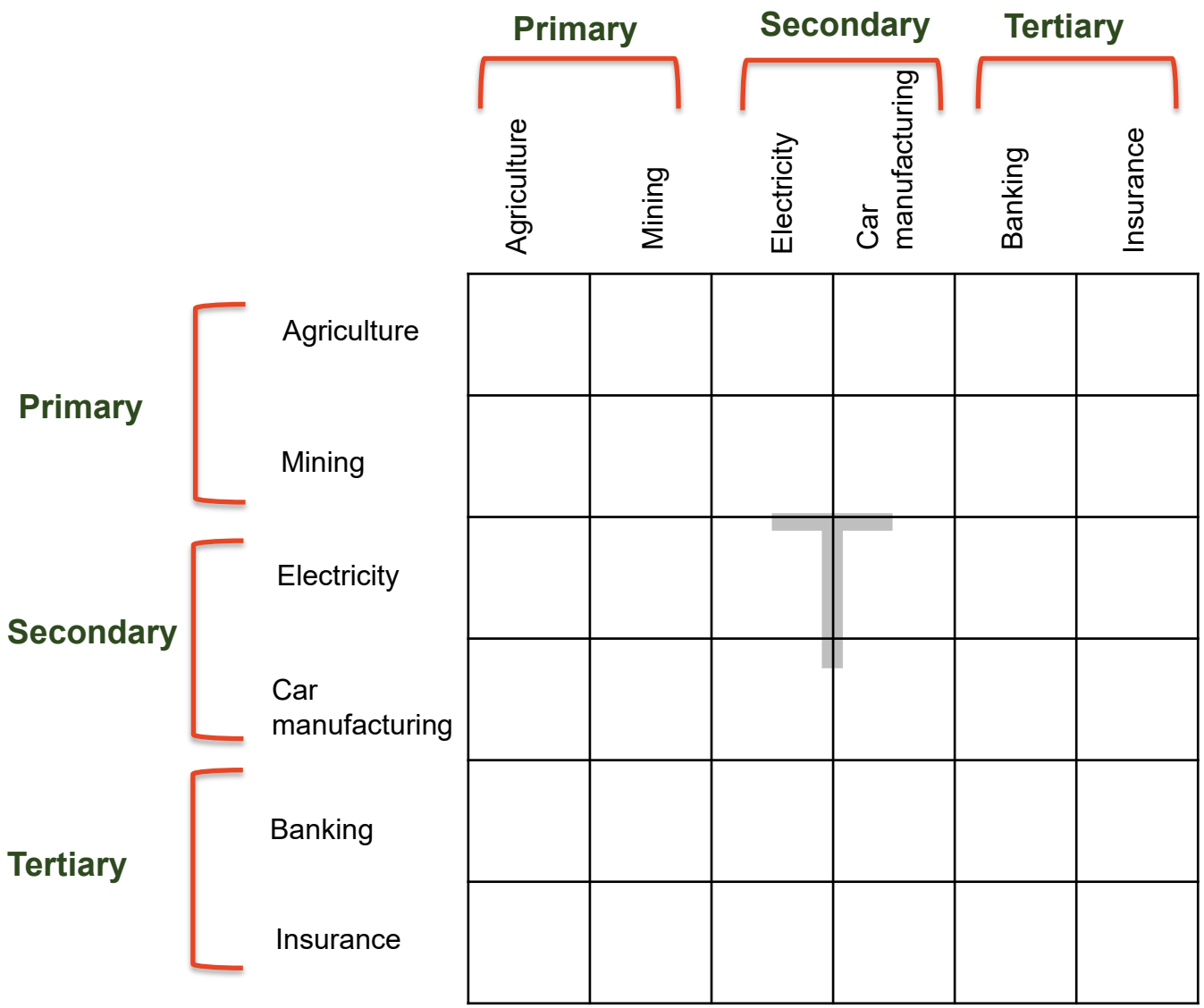
3 matrices:

Intermediate demand
Final demand
Value-added



3 matrices:

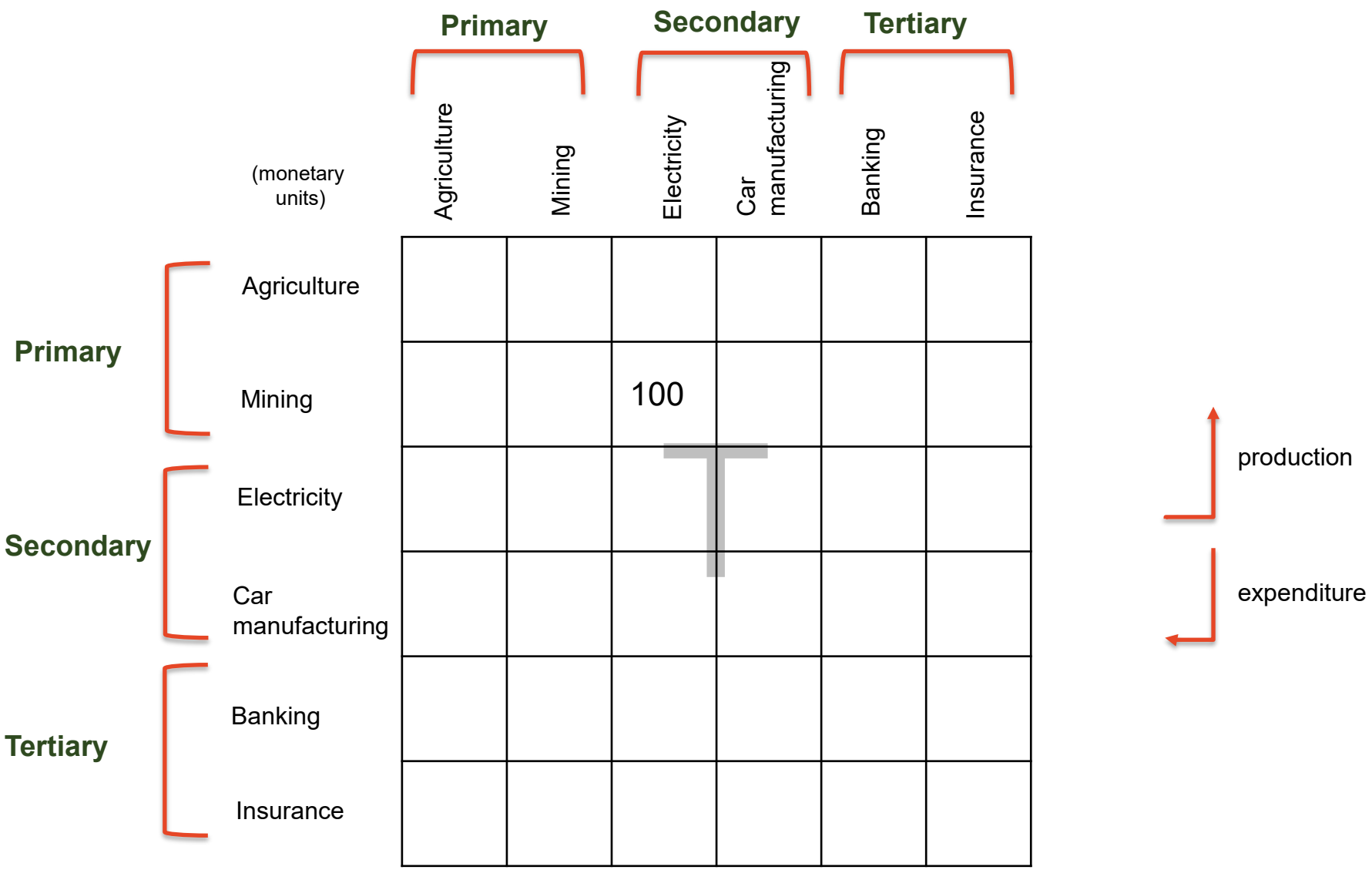
Intermediate demand
Final demand
Value-added



Sector to sector transactions

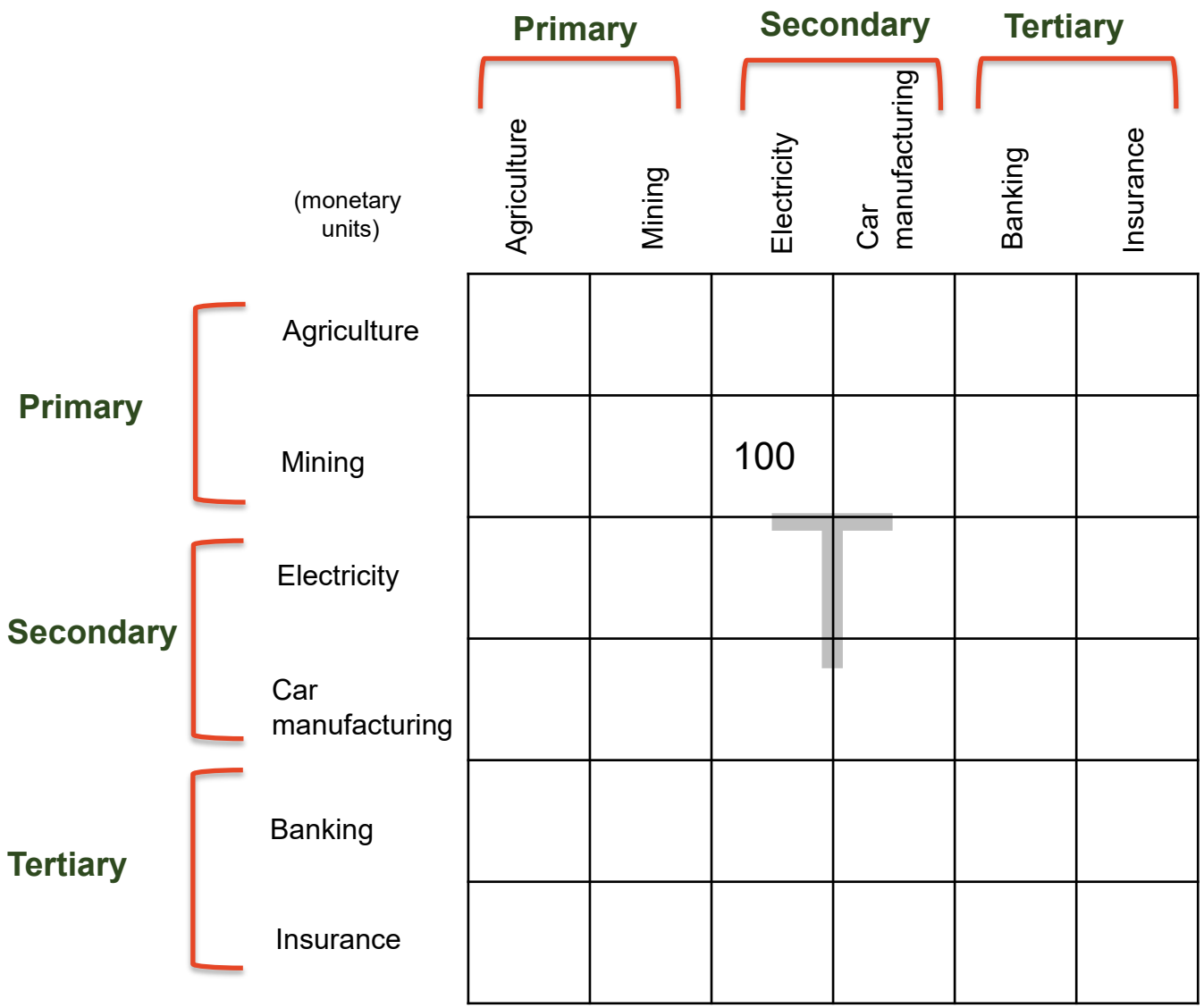
Intermediate demand (T)

The matrix that records the value of the flow of goods and services between sectors, with the inputs subsequently used to produce the output of the purchasing sector.



Intermediate demand (T)

The matrix that records the value of the flow of goods and services between sectors, with the inputs subsequently used to produce the output of the purchasing sector.



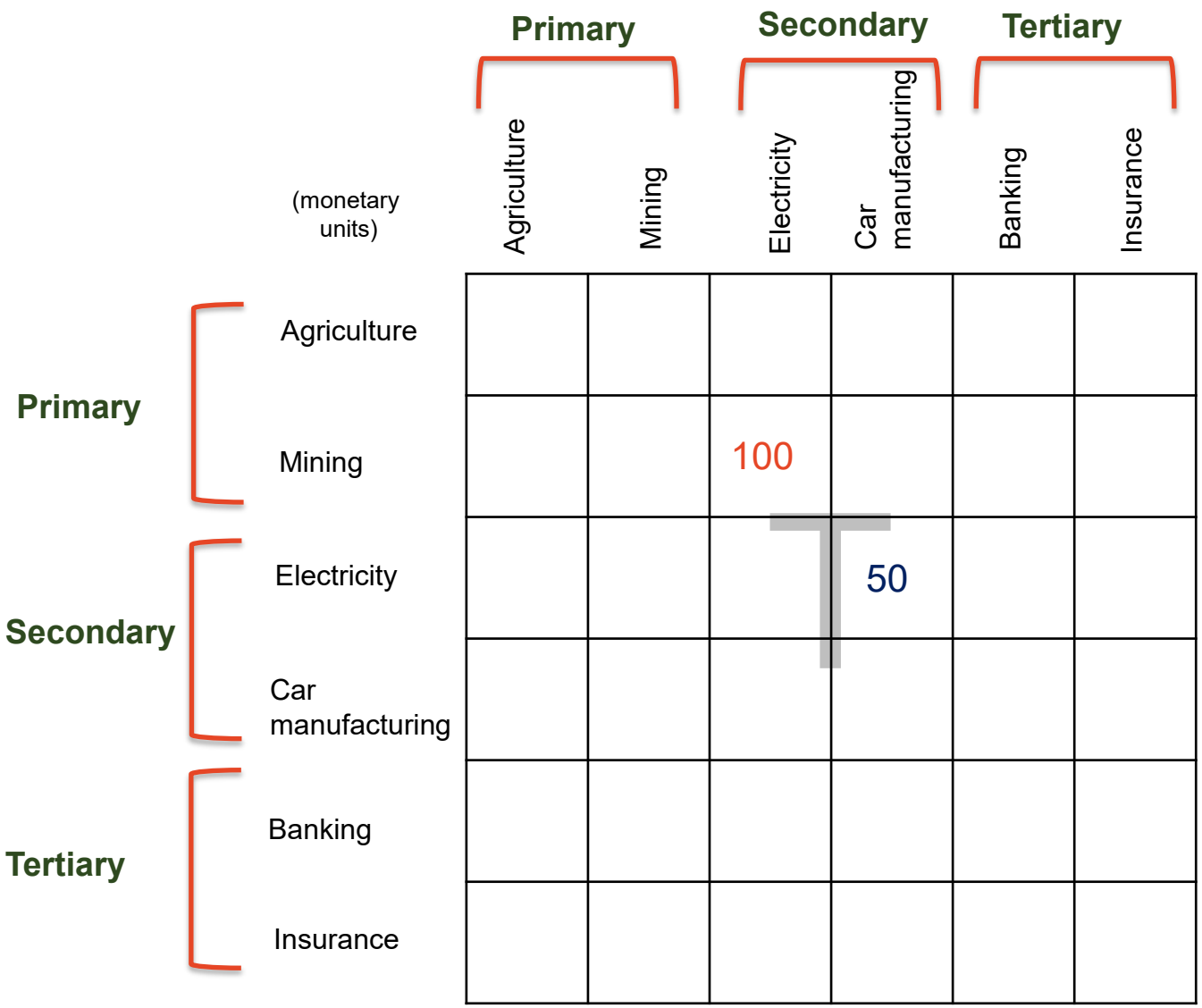
The electricity sector uses $100 worth of inputs (goods) from the mining sector (e.g. coal)

production

expenditure

Intermediate demand (T)

The matrix that records the value of the flow of goods and services between sectors, with the inputs subsequently used to produce the output of the purchasing sector.



The electricity sector uses $100 worth of inputs (goods) from the mining sector (e.g. coal)

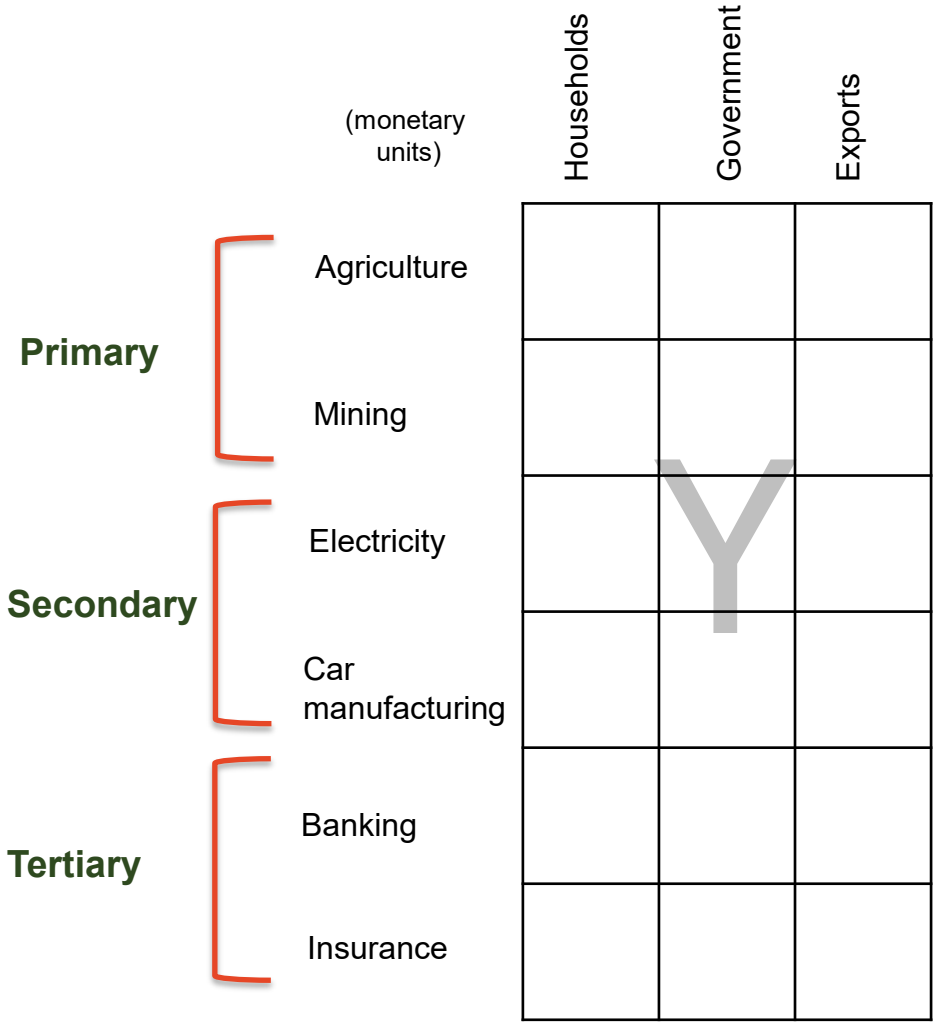
production

expenditure

The car manufacturing sector buys $50 worth of inputs from the electricity sector

Intermediate demand (T)

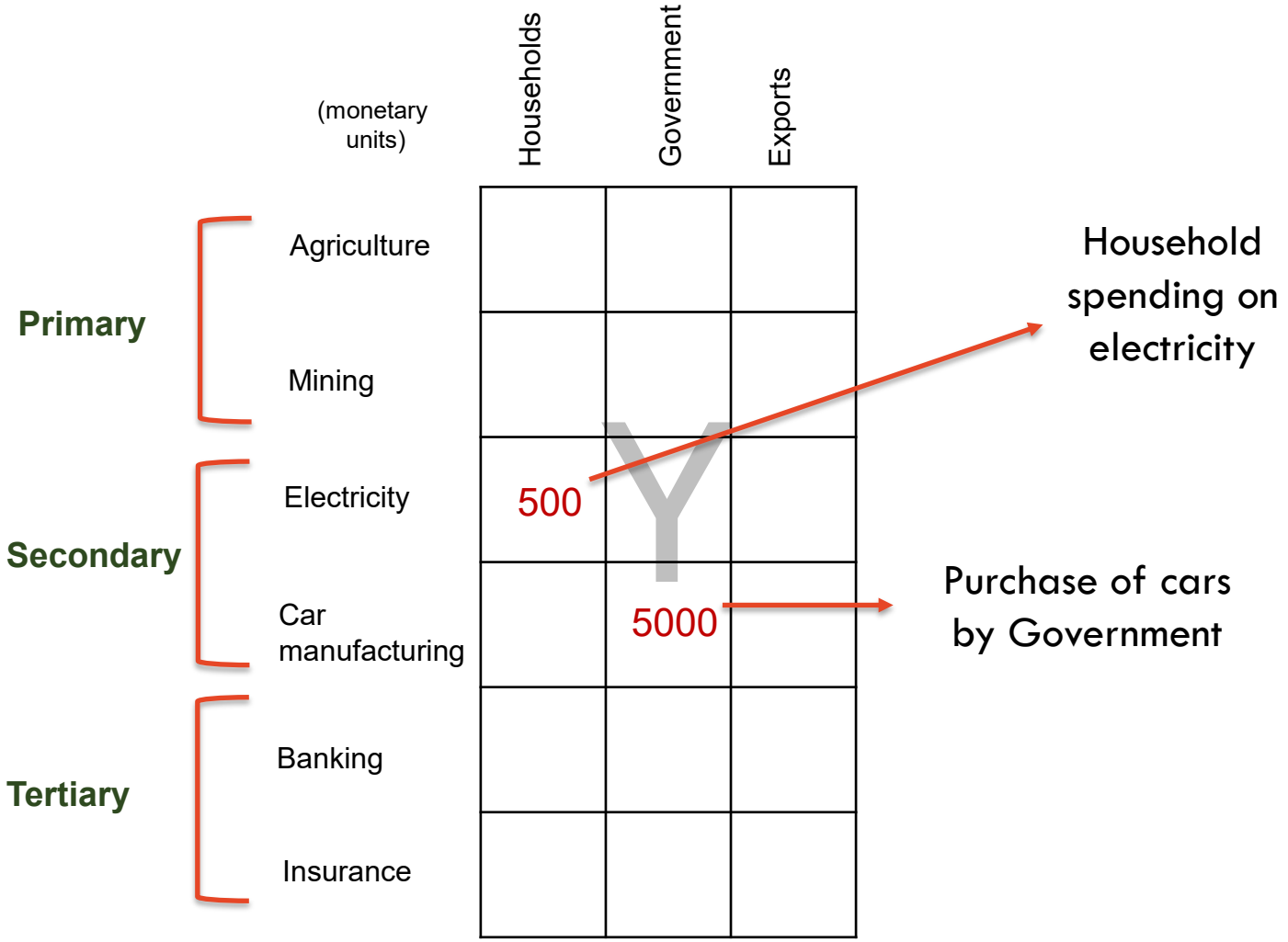
The matrix that records the value of the flow of goods and services between sectors, with the inputs subsequently used to produce the output of the purchasing sector.



Sector to consumer transactions

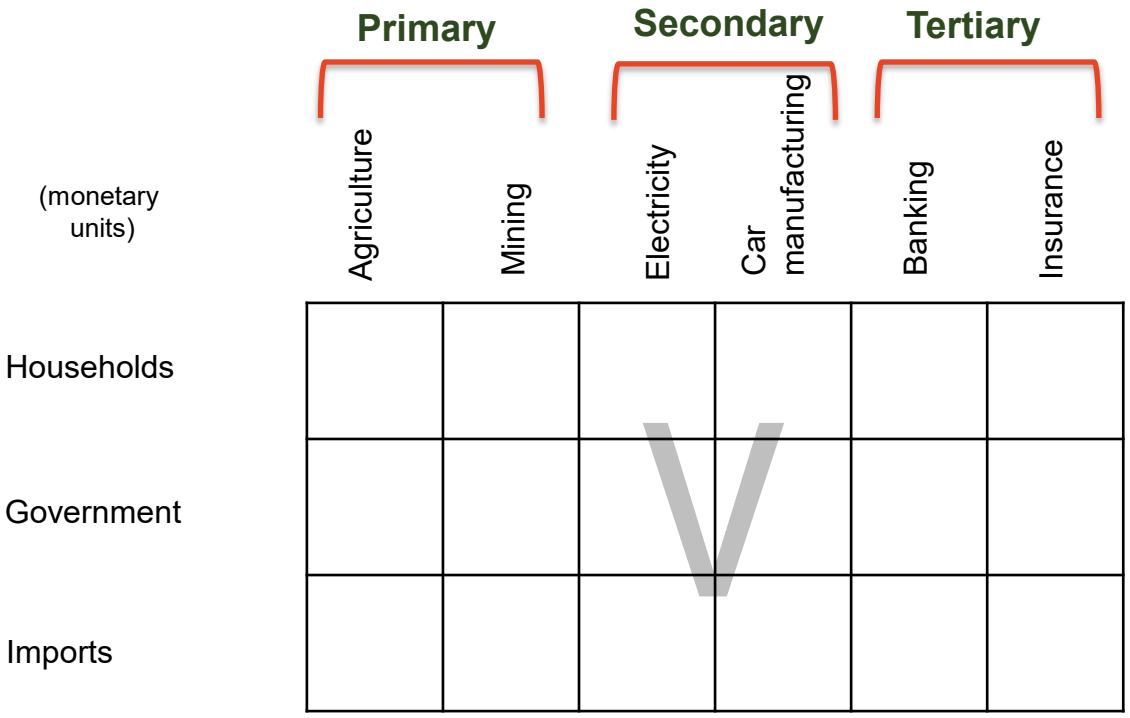
Final demand (Y)

The matrix that records the value of the final sale of goods and services to non-industry consumers



Final demand (Y)

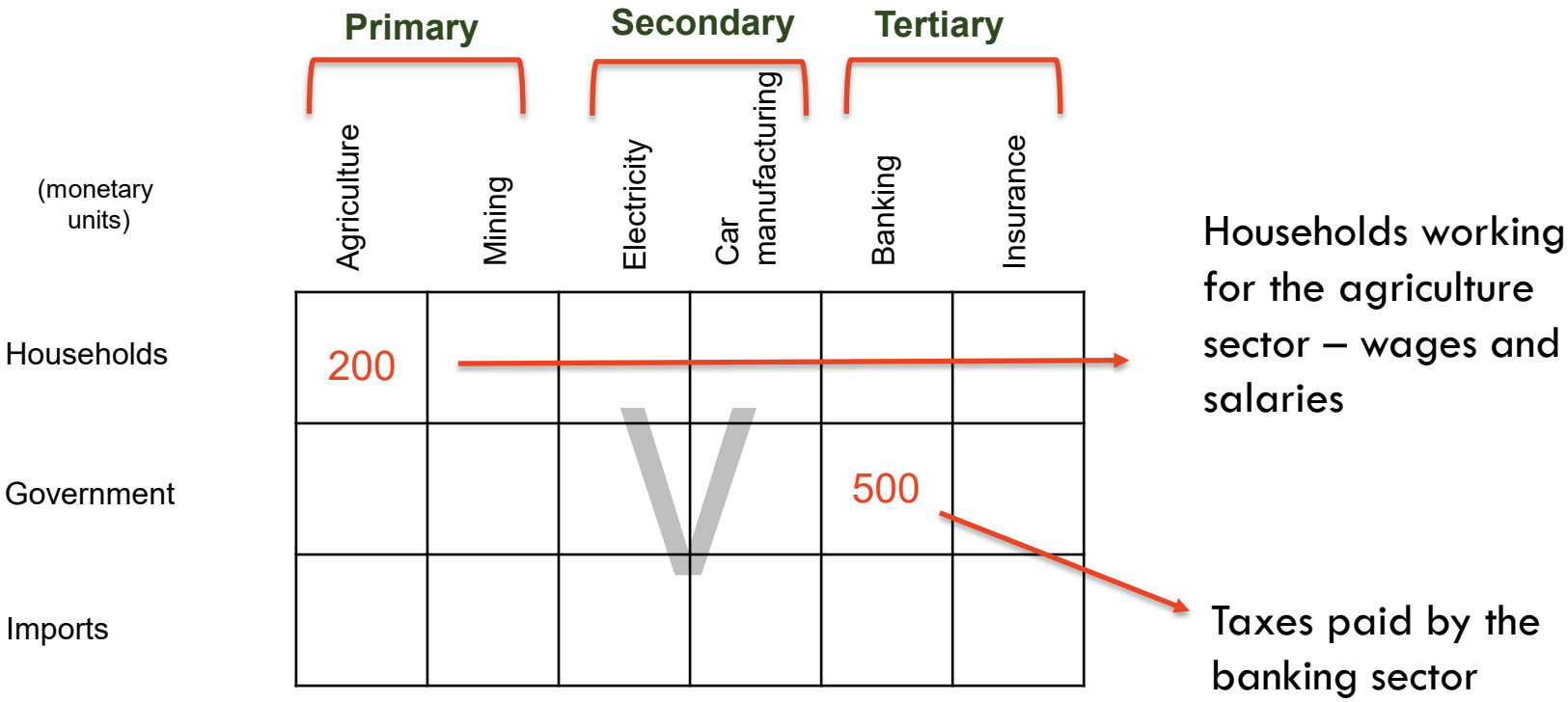
The matrix that records the value of the final sale of goods and services to non-industry consumers



All other inputs to production not transacted between sectors

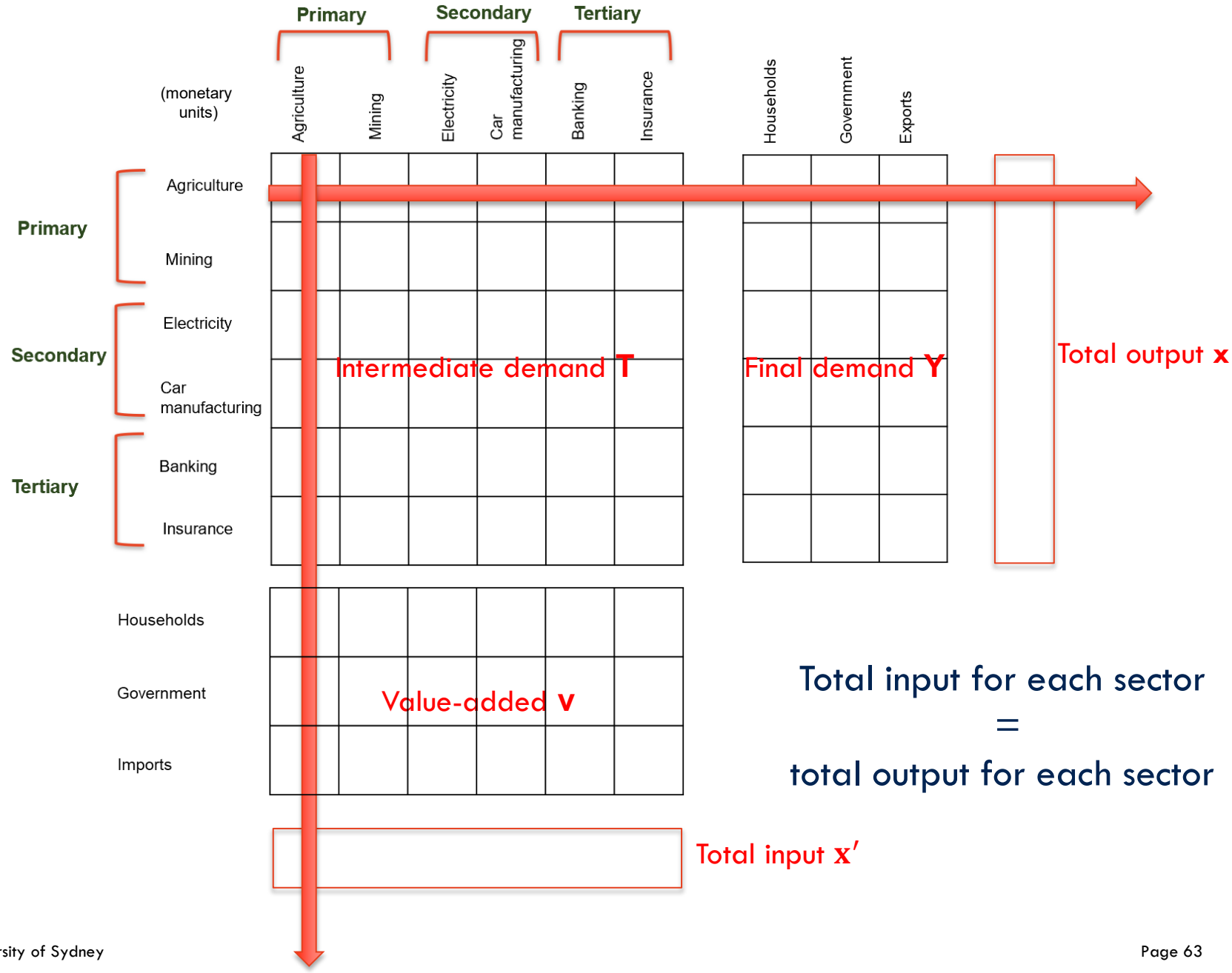
Value-added (v)

The matrix that records the value of the primary inputs needed to produce the goods and services of each sector



Value-added (v)

The matrix that records the value of the primary inputs needed to produce the goods and services of each sector

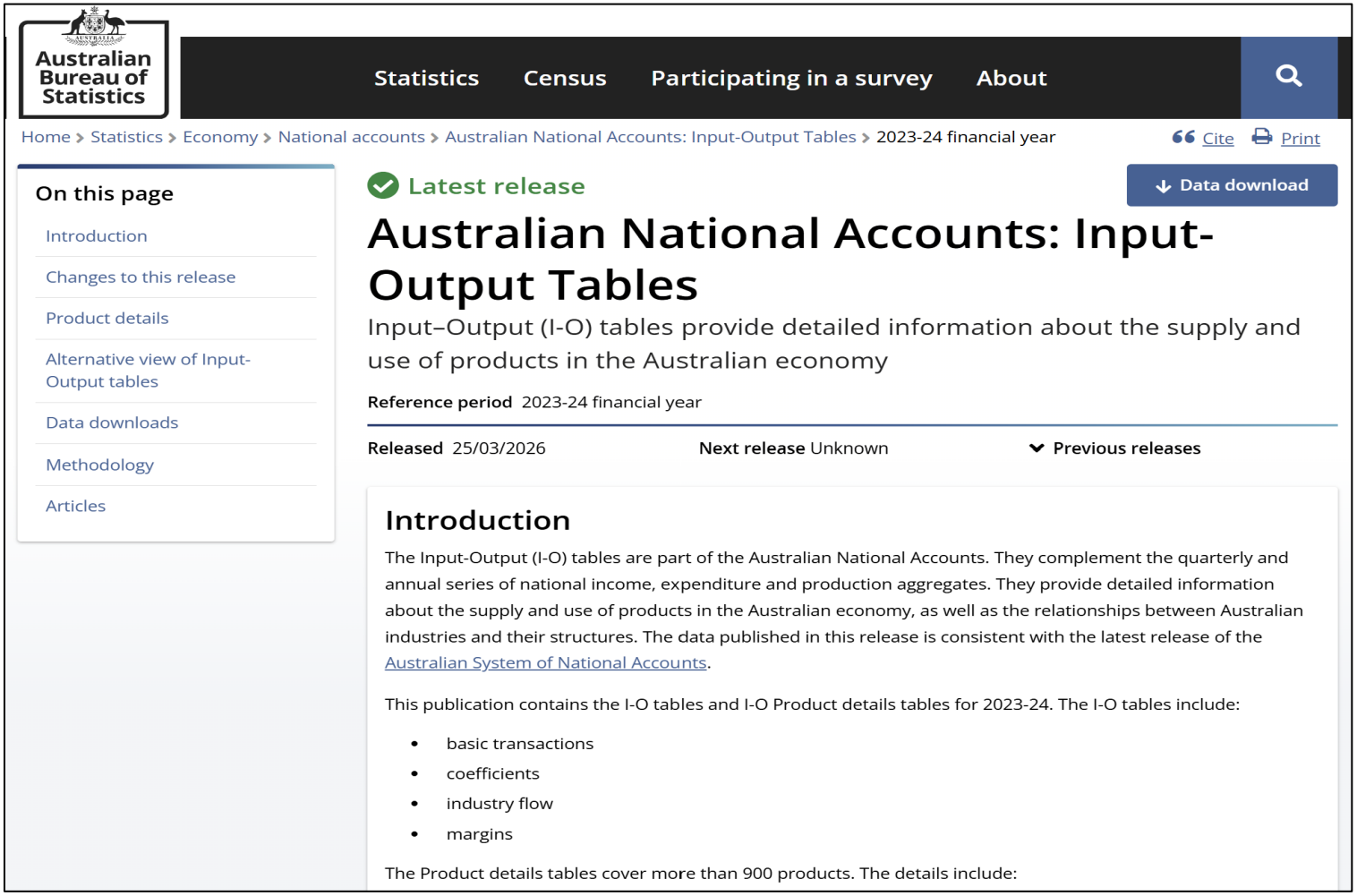


Total input for each sector

=

total output for each sector

Australian Bureau of Statistics



https://www.abs.gov.au/statistics/economy/national-accounts/australian-national-accounts-input-output-tables/latest-release

Australian Bureau of Statistics

Product details

This data presents information on supply and use by industry and product. It shows Australian production, imports, intermediate usage, final usage, exports, margins and taxes less subsidies on products.

Columns in the Input-Output (I-O) tables and I-O Product details tables both show the industry classification Input-Output Industry Group (IOIG). Rows in the I-O tables show the product classification Input-Output Product Group (IOPG). Rows in the I-O Product details tables show the product classification Input-Output Product Classification (IOPC).

IOIGs are based on the Australian and New Zealand Standard Industrial Classification (ANZSIC). IOPGs are groups of IOPCs aggregated to the IOIGs to which they are primary.

The IOPC is an industry-of-origin product classification that has been specifically developed for the Australian I-O tables. The I-O system describes the production and subsequent use of all goods and services. Each I-O product classification is defined by characteristic products of industry sectors. The overall principles for such an industry-of-origin product classification include:

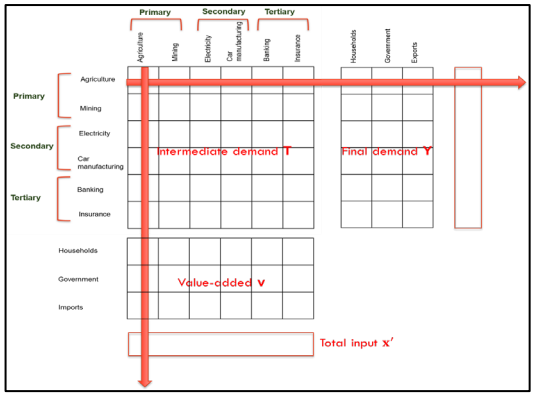
- Homogeneity of inputs - each product or product group should consist of items that have similar input structures or technology of production. This principle is generally applied through the definition of each IOPC item in terms of the ANZSIC industry in which it is primarily produced.
- Homogeneity of disposition - each product or product group, having satisfied the first criterion, should consist of items that have similar patterns of disposition or usage. This principle is applied by reference to the description of source data items and information about the transport, distribution and product taxation margins applying to particular products.

https://www.abs.gov.au/statistics/economy/national-accounts/australian-national-accounts-input-output-tables/latest-release

Sheep, Grains, Beef and Dairy Cattle
Poultry and Other Livestock
Other Agriculture
Aquaculture
Forestry and Logging
Fishing, hunting and trapping
Agriculture, Forestry and Fishing Support Services
Coal Mining
Oil and Gas Extraction
Iron Ore Mining
Non Ferrous Metal Ore Mining
Non Metallic Mineral Mining
Exploration and Mining Support Services
Meat and Meat Product Manufacturing
Processed Seafood Manufacturing
Dairy Product Manufacturing
Fruit and Vegetable Product Manufacturing
Oils and Fats Manufacturing
Grain Mill and Cereal Product Manufacturing
Bakery Product Manufacturing
Sugar and Confectionery Manufacturing
Other Food Product Manufacturing
Soft Drinks, Cordials and Syrup Manufacturing
Beer Manufacturing
Wine, Spirits and Tobacco
Textile Manufacturing
Tanned Leather, Dressed Fur and Leather Product Manufacturing
Textile Product Manufacturing
Knitted Product Manufacturing
Clothing Manufacturing
Footwear Manufacturing
Sawmill Product Manufacturing

Footwear Manufacturing
Sawmill Product Manufacturing
Other Wood Product Manufacturing
Pulp, Paper and Paperboard Manufacturing
Paper Stationery and Other Converted Paper Product Manufacturing
Printing (including the Reproduction of Recorded Media)
Petroleum and Coal Product Manufacturing
Human Pharmaceutical and Medicinal Product Manufacturing
Veterinary Pharmaceutical and Medicinal Product Manufacturing
Basic Chemical Manufacturing
Cleaning Compounds and Toiletry Preparation Manufacturing
Polymer Product Manufacturing
Natural Rubber Product Manufacturing
Glass and Glass Product Manufacturing
Ceramic Product Manufacturing
Cement, Lime and Ready-Mixed Concrete Manufacturing
Plaster and Concrete Product Manufacturing
Other Non-Metallic Mineral Product Manufacturing
Iron and Steel Manufacturing
Basic Non-Ferrous Metal Manufacturing
Forged Iron and Steel Product Manufacturing
Structural Metal Product Manufacturing
Metal Containers and Other Sheet Metal Product Manufacturing
Other Fabricated Metal Product Manufacturing
Motor Vehicles and Parts; Other Transport Equipment manufacturing
Ships and Boat Manufacturing
Railway Rolling Stock Manufacturing
Aircraft Manufacturing
Professional, Scientific, Computer and Electronic Equipment Manufacturing
Electrical Equipment Manufacturing
Domestic Appliance Manufacturing
Specialised and Other Machinery and Equipment Manufacturing

PHYS5033 journey



Footprints calculations: bringing it all together !

$$\mathbf{x} = (\mathbf{I} - \mathbf{A})^{-1}\boldsymbol{y}$$

$$\mathbf{x} = \mathbf{L}\boldsymbol{y}$$

Multiply both sides by $\mathbf{q}$

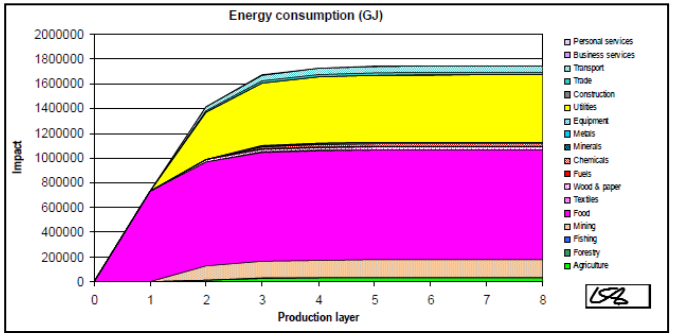
$$\mathbf{q}\mathbf{x} = \mathbf{q}\mathbf{L}\boldsymbol{y}$$

$$\mathbf{q}\mathbf{x} = \mathbf{m}\boldsymbol{y}$$

Direct intensities

$$\boldsymbol{Footprint} = \mathbf{m}\boldsymbol{y} = \mathbf{q}\mathbf{L}\boldsymbol{y}$$

Total intensities



TUTORIAL ACTIVITIES



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Tutorial Activities - I
Reading an input-output table

Tutorial Activities

- **What does an input-output table look like?** scan through the various elements of the following input-output table - download IO table

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	Sample unbalanced IO table	Unit: USD						Australia	Australia	Australia	Australia	Australia	Australia	Australia	Australia	Australia	Australia	Australia	Australia
2					AUS	AUS	AUS	AUS	AUS	AUS	AUS	AUS	AUS	AUS	AUS	AUS	AUS	AUS	AUS
3					Industries	Industries	Industries	Industries	Industries	Industries	Industries	Industries	Industries	Industries	Industries	Industries	Industries	Industries	Industries
4					Sheep and l	Shorn woo	Oats, sorgl	Wheat	Barley	Rice	Oilseeds	Legumes	Beef cattle	Untreated	Dairy cattl	Pigs	Poultry	Eggs	Vegetables
5	Australia	AUS		Sheep and lambs	8.32E+05	8.27E+05	1.12E+03	2.70E+02	4.06E+02	2.74E+02	2.82E+02	3.00E+02	6.86E+02	5.40E+02	5.11E+02	7.33E+02	8.44E+02	5.39E+02	1.78E+06
6	Australia	AUS		Shorn wool	7.00E+03	2.32E+03	2.72E+03	6.58E+02	1.31E+03	9.20E+02	8.83E+02	9.53E+02	4.04E+03	4.10E+03	5.01E+03	2.16E+04	2.26E+03	6.82E+02	2.10E+03
7	Australia	AUS		Oats, sorghum and other cereal g	1.13E+06	1.02E+05	1.12E+08	1.53E+02	2.61E+02	2.61E+02	2.16E+02	2.70E+02	1.25E+06	1.99E+05	2.02E+05	1.80E+06	3.28E+06	4.63E+02	2.67E+05
8	Australia	AUS		Wheat	4.67E+07	8.52E+06	2.09E+03	9.07E+08	1.41E+03	1.49E+03	1.21E+03	1.55E+03	3.32E+07	4.70E+06	4.06E+06	5.70E+06	8.74E+06	2.50E+03	1.48E+07
9	Australia	AUS		Barley	2.20E+06	3.84E+05	7.54E+02	2.63E+02	2.52E+07	4.81E+02	3.89E+02	4.97E+02	4.73E+06	7.73E+05	4.17E+05	4.07E+06	7.70E+06	8.12E+02	8.31E+04
10	Australia	AUS		Rice	5.88E+02	5.59E+02	1.05E+03	3.04E+02	4.95E+02	2.28E+06	2.93E+02	3.73E+02	1.67E+03	1.67E+03	1.31E+03	6.09E+02	1.78E+04	6.20E+02	2.52E+05
11	Australia	AUS		Oilseeds	1.61E+04	1.28E+04	1.87E+04	4.92E+03	7.72E+03	4.55E+03	3.06E+07	3.76E+03	4.61E+06	1.15E+04	1.12E+04	1.66E+04	1.68E+04	5.75E+03	1.95E+04
12	Australia	AUS		Legumes	1.07E+07	3.28E+06	5.12E+03	1.09E+03	1.98E+03	1.49E+03	1.47E+03	7.48E+06	3.57E+07	2.22E+06	1.54E+06	6.18E+06	5.89E+06	3.16E+05	1.10E+07
13	Australia	AUS		Beef cattle	5.67E+02	5.57E+02	7.31E+02	2.01E+02	2.82E+02	2.23E+02	2.38E+02	2.57E+02	9.27E+06	8.92E+05	5.24E+05	5.32E+02	6.22E+02	5.07E+02	7.02E+05
14	Australia	AUS		Untreated milk	4.25E+03	3.76E+03	5.22E+03	1.24E+03	2.24E+03	1.38E+03	1.45E+03	1.70E+03	2.20E+07	4.12E+06	1.61E+04	3.75E+03	4.13E+03	1.01E+03	6.86E+05
15	Australia	AUS		Dairy cattle	1.25E+02	1.08E+02	1.08E+02	9.86E+01	9.77E+01	1.03E+02	9.82E+01	1.01E+02	1.28E+02	1.46E+02	1.32E+02	1.58E+02	1.78E+02	3.01E+02	1.45E+02
16	Australia	AUS		Pigs	4.93E+02	4.88E+02	5.98E+02	1.48E+02	2.20E+02	1.96E+02	1.97E+02	2.10E+02	4.41E+02	3.96E+02	3.17E+02	2.56E+02	3.05E+02	1.14E+03	1.22E+03
17	Australia	AUS		Poultry	5.36E+02	5.45E+02	6.38E+02	1.75E+02	2.48E+02	2.03E+02	2.21E+02	2.33E+02	4.78E+02	4.34E+02	3.49E+02	2.80E+02	1.00E+07	3.94E+06	1.34E+03
18	Australia	AUS		Eggs	1.09E+03	1.14E+03	1.29E+03	3.11E+02	4.81E+02	3.05E+02	3.90E+02	4.10E+02	9.77E+02	8.27E+02	6.27E+02	3.48E+02	1.04E+03	3.30E+02	6.87E+02
19	Australia	AUS		Vegetables	8.89E+05	7.54E+05	5.14E+02	1.57E+02	2.33E+02	2.33E+02	1.94E+02	2.25E+02	4.69E+06	7.66E+04	1.12E+05	1.48E+05	2.28E+02	3.83E+02	1.58E+07
20	Australia	AUS		Fruit	3.19E+02	3.15E+02	2.80E+02	1.45E+02	2.12E+02	2.15E+02	1.88E+02	1.98E+02	3.20E+02	3.35E+02	3.04E+02	2.21E+02	2.20E+02	3.43E+02	1.85E+02
21	Australia	AUS		Plant nurseries	2.13E+02	1.95E+02	1.96E+02	1.39E+02	1.69E+02	2.07E+02	1.80E+02	1.90E+02	2.23E+02	2.42E+02	2.47E+02	2.13E+02	2.12E+02	2.31E+02	2.24E+02
22	Australia	AUS		Flowers	2.80E+02	2.89E+02	3.17E+02	1.11E+02	1.59E+02	1.66E+02	1.44E+02	1.55E+02	2.82E+02	2.65E+02	2.27E+02	1.71E+02	1.70E+02	2.45E+02	7.53E+02
23	Australia	AUS		Grapes for wine	4.03E+03	3.97E+03	4.97E+03	1.21E+03	2.02E+03	1.77E+03	1.46E+03	1.69E+03	3.29E+03	2.63E+03	2.19E+03	1.51E+03	1.74E+03	2.34E+03	2.71E+03
24	Australia	AUS		Horses	9.24E+02	8.93E+02	1.37E+03	3.37E+02	5.06E+02	3.41E+02	3.51E+02	3.55E+02	8.48E+02	6.58E+02	6.56E+02	5.50E+02	6.33E+02	9.90E+02	6.49E+04
25	Australia	AUS		Deer	6.43E+02	6.84E+02	8.63E+02	2.14E+02	3.12E+02	1.95E+02	2.44E+02	2.56E+02	5.86E+02	4.49E+02	3.53E+02	3.30E+02	3.98E+02	9.69E+02	3.20E+04
26	Australia	AUS		Animal breeding	5.21E+02	5.41E+02	7.10E+02	1.80E+02	2.61E+02	1.81E+02	2.18E+02	2.30E+02	4.80E+02	4.10E+02	3.22E+02	2.75E+02	3.26E+02	8.15E+02	1.82E+04
27	Australia	AUS		Sugar cane	1.63E+04	9.49E+03	1.22E+04	6.70E+03	1.30E+04	1.26E+04	9.01E+03	1.17E+04	1.24E+04	1.90E+04	1.46E+04	9.04E+03	7.92E+03	4.27E+03	8.17E+03
28	Australia	AUS		Unginned cotton	5.33E+02	5.67E+02	7.07E+02	2.21E+02	3.09E+02	2.65E+02	2.67E+02	2.70E+02	4.90E+02	5.01E+02	4.32E+02	2.56E+02	3.02E+02	3.03E+02	4.09E+02
29	Australia	AUS		Hops	5.86E+02	4.82E+02	6.20E+02	3.04E+02	4.41E+02	4.82E+02	3.56E+02	4.21E+02	5.84E+02	7.78E+02	6.42E+02	3.37E+02	3.37E+02	7.42E+02	3.81E+02
30	Australia	AUS		Grass seed	2.98E+06	1.74E+06	2.08E+02	1.47E+02	1.80E+02	2.24E+02	1.93E+02	2.03E+02	9.01E+06	1.63E+06	9.07E+05	7.70E+04	2.24E+02	1.83E+02	1.23E+03
31	Australia	AUS		Hay	4.48E+07	3.55E+07	1.47E+06	4.41E+05	1.42E+06	7.37E+05	7.23E+05	9.53E+05	1.27E+08	2.37E+07	7.20E+06	1.84E+07	3.20E+06	1.25E+06	1.19E+03
32	Australia	AUS		Natural rubber	2.10E+02	1.84E+02	2.03E+02	1.28E+02	1.68E+02	1.95E+02	1.58E+02	1.73E+02	2.18E+02	2.64E+02	2.39E+02	2.06E+02	1.97E+02	1.28E+02	1.38E+02
33	Australia	AUS		Skins and other agricultural servic	7.12E+03	4.55E+03	4.20E+04	9.67E+04	2.40E+04	6.02E+03	4.52E+03	7.22E+03	1.92E+07	2.87E+07	1.63E+06	1.04E+06	2.54E+06	1.55E+05	1.03E+07

IO_table

This number format is scientific notation, where 2.10E+02 represents 2.10×10^2, which equals 210.

Tutorial Activities – II
Make your own input-output table



Tutorial Activities

- **Construct an IO table:** Build your own imaginary input-output table for an economic system of your choice. Your input-output table should have the following specifications:

    1. 14 intermediate demand (industry) sectors
    2. 3 value-added (primary inputs) entities (Households, Government, Imports)
    3. 3 final demand entities (Households, Government, Exports)
    4. Add numbers to your input-output table. For adding numbers, you can refer to published input-output tables that are available online. If you are making a dummy input-output table, the numbers can be made-up as well. The important aspect is that your structure should be correct! The purpose of the exercise is to construct an input-output table and use it for learning the techniques for footprint assessments.

https://canvas.sydney.edu.au/courses/74558/pages/week-1-introduction-and-structure-of-input-output-tables

How to make an input-output table?

Two possible options:

Option 1: Take an existing input-output table and aggregate the sectors to yield a 14 x 214 sector intermediate demand matrix.

Option 2: Construct your own input-output table after reviewing a sample table. Select 14 sectors and fill in some numbers. Ensure that the structure makes sense, i.e. 14 sectors in rows in the same order as the 14 sectors in columns.

Try to include a full range of sectors so that all economic activity is covered by your table, since this is how real input-output tables are structured.

Homework task assignment

Homework week 1 (Due: September 9, 2026)

Build your own hypothetical input-output table for an economic system of your choice.

14 industry sector input-output table

Make sure that your table features:

- 14 intermediate demand (industry) sectors
- 3 value-added (primary inputs) entities (Households, Government, Imports)
- 3 final demand entities (Households, Government, Exports)
- Add numbers to your input-output table. For adding numbers, you can refer to published input-output tables that are available online, or you can just make the numbers up. The important aspect is that your structure should be correct!

The input-output system needs to be presented in monetary units – it is up to you which currency you want your input-output table to be in, but you must include the relevant units when presenting the table.

EXPLANATION: Explain the various components of an input-output table - intermediate demand, final demand and value-added matrix (about 150 words).

Please build your input-output table in Excel. You will need this input-output table for the remainder of this course.

Homework task assignment tips #1

- Start a new Excel file
- Make 10 tabs – label them: Week 1, Week 2, …, Week 10
- Start with Week 1
- Read the instructions on Canvas
- Complete the Homework task
- Save the file
- Use the same file for Weeks 2 – 10
- You can add text boxes to help 'future you' remember what you did
- Include calculations for all questions specified in the Homework Task assignment – i.e. show your working. You can use formulae in Excel to help you with this.
- Answer the explanation questions each week in a separate Word file (this needs to be uploaded separately to the Excel file)
- This is a resource for you!

PHYS5033 organisation – assessment tasks

Tips for the Homework tasks:

- Make sure that you read the homework instructions thoroughly for each week – if you don't complete one of the steps expected, you won't get any marks for it.
- If there is a question posed in one of the homework steps, make sure you provide an explicit answer for that question in the Excel file.
- Always include labels and units whenever you present results and make sure that your workings are included.
- You will be provided with worked examples and tutorial answers. These provide you with examples of how the calculations work – they do not provide you with a template for the homework. If you just replicate these, you will not be answering the steps expected in the homework.
- You should be able to answer the questions posed in the 'Explanation' part of the homework each week using the content provided in the lectures and in the background reading material. There is no need to use other sources to answer these questions, and doing so often results in students providing irrelevant or incorrect answers. If you do use other sources of information, these sources must be cited and referenced appropriately. You are strongly advised NOT to rely on AI tools to source your answers for these questions since they usually provide incorrect information.
- Make sure that your Excel submission is easy for the marker to read and find your results. This includes using appropriate font sizes, using appropriate row heights and column widths, and including text to note which step the results you're presenting relate to.

Key takeaways

- What are input-output tables?
- What is the structure of input-output tables?
- What is the difference between the intermediate demand, final demand and value-added matrices?
- General feel of the type of footprint assessments that can be done using input-output tables

More to come:

- Matrix fun!
- Footprint assessments